SMART INDIA HACKATHON / SOFTWARE / AICTE REGULATION BUREAU

NEXUS-IC

COMPLETE SOLUTION & IMPLEMENTATION BLUEPRINT

Problem Statement 26264

Integrated Dashboard for Monitoring and Tracking
Institution-Related Complaint Cases Across AICTE Bureaus/Cells

NEXUS SYNC - INTELLIGENT CROSS-BUREAU COORDINATION

NEXUS PREVENT + NEXUS SYNC: prevent duplicate action, coordinate safely, verify outcomes.

64 baseline features + 10 integrated innovations + 9 verified corrections | 23 OCTOBER 2026

Implementation proposal - not a deployed AICTE system. Synthetic data assumed; official integrations, legal timelines, authority and pilot outcomes must be verified with AICTE.

0. Read this specification first

What is being built	An internal case-intelligence and coordination application for authorized AICTE officers. It does not replace official grievance intake, does not automatically decide case guilt or status, and never claims live AICTE/CPGRAMS access during a synthetic prototype.
Primary innovation	NEXUS PREVENT checks proposed administrative actions before work is duplicated. NEXUS SYNC: identity-safe case overlap review -> evidence explanation -> blind bureau-to-bureau notice -> authorized acknowledgement -> lead and task assignment -> outcome receipt. Case source truth remains intact.
Engineering baseline	Frontend: React with JSX and JavaScript (Vite); no TypeScript. Backend: Express with plain JavaScript on Node.js. Database: MongoDB/Mongoose. UI: Tailwind, React Router, TanStack Query, Recharts; optional React Flow for the graph. Test: Vitest, React Testing Library, Supertest, Playwright.
Document contract	P0 denotes essential implemented-and-demonstrated behavior; P1 adds competitive proof; P2 is planned/optional. Every item is proposed until actual evidence exists. All visualizations and statements use only synthetic institutions unless authorized data is later supplied.

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Section	Purpose
19	Additional enhancement tests
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1. Product purpose and constraints

AICTE receives institution-related grievances that may be processed by RB, PGRC and other Bureaus/Cells. The stated challenge is the inability to see and coordinate all matters concerning one institution, creating potential repeated work, untracked ownership and slow disposal. NEXUS-IC consolidates authorized read views and adds a tracked coordination response. [P]

What the product is - and is not

Is	Is not
Internal institution-wise coordination and monitoring layer	Replacement for existing complaint portals or a promise of CPGRAMS write access
Human-verified, explainable potential overlap detector	Automatic merger of source cases; wrongdoing or legality classifier
Scoped workflow with acknowledgements and audit trail	Unapproved bureau disclosure or automated statutory decisions
Configurable policy-specific SLA monitor	Assumed single AICTE-wide 21-day legal rule
Synthetic-data demonstrator with reproducible tests	AICTE deployment, statutory validation or measured operational savings

Objective hierarchy

1. Prove all PS 26264 functional requirements are represented by a working, persistent screen or service.

2. Prove one end-to-end RB-PGRC coordination interaction is safe and accountable.

3. Prove hard negatives: similar names, different incidents, different proceedings, stale feeds, denied access.

4. Measure technical quality and usability; do not invent review precision, time savings or successful rollout.

Priority contract

Tier	Meaning	Evidence for sign-off
P0	Core SIH demonstration and PS compliance	Working backend mutation, persistence, integration or E2E test, and reviewer screenshot.
P1	Strong finals differentiator after P0 is stable	Reviewer-visible interaction plus positive and negative test.
P2	Optional post-core enhancement	Feature flagged and clearly labeled; cannot masquerade as built.

Before versus after - fictional pain story, not measured impact

Conventional fragmented handling	NEXUS-IC with authorized cooperation
Monday: PGRC receives a complaint about unreturned certificates. Its own case PG-031 is opened.	Monday: PG-031 is stored with verified AICTE Permanent ID and native PGRC state.
Tuesday: RB gets a differently phrased complaint from the same institution and opens RB-008.	RB-008 is retained as an independent case. Identity gate and issue matching flag a potential relation.

Conventional fragmented handling	NEXUS-IC with authorized cooperation
Officers independently draft the same request for academic documents.	NEXUS PREVENT checks an action proposal and returns REVIEW with linked reasons.
Two request letters may be issued; neither bureau knows the other action until later.	Officer reviews the exact evidence; the banner invites NEXUS SYNC, with no automatic merger.
No single acknowledgement or shared next-step owner is visible.	PGRC accepts a blind coordination request; authorized lead, tasks, dates and receipt are recorded.
Duplicated effort is plausible, but no real savings have been established.	If one redundant letter is explicitly cancelled and recorded, the synthetic trial counts one demonstrated avoidance.

Before/after quantities are illustrative and must not be presented as savings achieved at AICTE. Impact Lab will run paired seeded simulations using identical fixtures and publish the raw event trace, denominators and fixture hash.

2. Exhaustive problem statement traceability

Every explicit sub-requirement from the supplied PS is mapped to one or more feature IDs. Inclusion here means included in the build specification, not already implemented. The acceptance test for each feature appears in Section 3.

R1 Institution profile

PS ID	Exact capability / interpretation	Planned feature IDs
R01	Unique AICTE Permanent ID	F01,F02
R02	Institution name, address, region, relevant metadata	F01,F04
R03	Consolidated institution complaint history	F04,F05,F33
R04	Identity and alias/campus ambiguity reviewed	F02,F03,F17

R2 Complaint tracking

PS ID	Exact capability / interpretation	Planned feature IDs
R05	Unique case/complaint ID and source ID	F07,F08
R06	Receipt date and complaint category	F08
R07	Native and current case status	F08,F09
R08	Present handling bureau/cell	F08,F10
R09	Assigned officer/official	F08,F10
R10	Last action and timestamp	F08,F11
R11	Next action required and expected disposal date	F08,F22,F39
R12	Transfer/escalation status, acknowledgement	F12,F41
R13	Remarks and documents linked	F11,F13

R3 Overlap detection

PS ID	Exact capability / interpretation	Planned feature IDs
R14	Recognize same institution handled by RB and PGRC	F17,F18
R15	Warn authorized officers about overlap	F20,F42

PS ID	Exact capability / interpretation	Planned feature IDs
R16	Link related cases without destroying source IDs	F14,F19
R17	Differently worded complaints considered	F18,F28
R18	Related-but-separate and false match supported	F18,F19,F14
R19	Prevent duplicated administrative coordination effort	F21,F22,F23

R4 Dashboard

PS ID	Exact capability / interpretation	Planned feature IDs
R20	Total complaint volume with stated scope	F29,F30
R21	Pending and under process counts	F30
R22	Disposed, transferred and escalated counts	F30
R23	Beyond-timeline and approaching due counts	F30,F40
R24	Institution-wise status visibility	F33,F04
R25	Bureau/cell-wise workload	F31
R26	As-of time and missing source not counted as zero	F29,F48

R5 Workflow and history

PS ID	Exact capability / interpretation	Planned feature IDs
R27	Complete chronological case actions	F11
R28	Officer/bureau responsible for event	F11,F47
R29	Date/time, remarks, file evidence	F11,F13
R30	Transfers and escalations with acknowledgements	F12,F41
R31	Coordinated handling and responsibilities	F21,F22,F26

R6 Search

PS ID	Exact capability / interpretation	Planned feature IDs
R32	Institution name / aliases / Permanent ID	F03,F32
R33	Complaint ID	F32
R34	Bureau/cell and region	F32
R35	Complaint category and case status	F32
R36	Receipt and action date-range filtering	F32
R37	Search restricted to role-allowed fields	F46

R7 Alerts

PS ID	Exact capability / interpretation	Planned feature IDs
R38	Potential duplicates and overlaps	F18,F20,F42
R39	Pending case reminders	F40

PS ID	Exact capability / interpretation	Planned feature IDs
R40	Escalation alerts	F41,F42
R41	Deadline approaching or breach alerts	F39,F40,F43
R42	Notification delivery/acknowledgement	F42,F44

R8 Reports

PS ID	Exact capability / interpretation	Planned feature IDs
R43	Bureau-wise and region-wise reports	F31,F34
R44	Institution-wise history	F04,F34
R45	Pending case ageing	F35
R46	Disposal rate and definitions	F35
R47	Overlapping/duplicate case trend reports	F36
R48	Monthly/quarterly analytics	F38
R49	Permission-checked export and audit	F34,F46,F47

Roles + outcomes + innovation

PS ID	Exact capability / interpretation	Planned feature IDs
R50	RB, PGRC, other bureaus, regional offices and managers	F45,F46
R51	Authorized senior management/competent authority views	F37,F46
R52	Rule-based or AI-assisted institution/incident matching	F17,F18,F28
R53	Coordination required alert and tracked outcome	F20,F21,F22,F26
R54	Institutional source-of-truth with reconciled disagreement	F48,F49
R55	Accountability, timely follow-up and data-driven decisions	F39,F40,F47,F35

Completeness check: 55 requirement checks covering institution data, every required case field, overlaps, dashboard status counts, movements, search filters, four alert categories, report categories, user roles and the matching innovation. Any new official PS amendment must produce a new traceability version.

Nine-gap review: correction locator

Gap	Correction designed in this edition	Validation
G01 - SLA defaults	Demo-only seeded timelines and exact trigger examples (Section 12)	D01-D05
G02 - matching vague	Formula, gates, thresholds, two calculated examples (Section 9)	D06-D12
G03 - banner absent	Exact quoted alert on dashboard, institution and review (Sections 8-9)	D13
G04 - Competent Authority	Dedicated role, dashboard, sign-off and policy controls (Section 10)	D14-D16
G05 - delivery channels	Email transport with retries; SMS sandbox/approval gate (Section 12)	D17-D19
G06 - single-window tension	Discoverable cross-bureau metadata; role-scoped narratives (Section 10)	D20-D23
G07 - no wireframes	Six black-and-white screen wireframes (Section 8)	D24
G08 - before-after missing	Fictional workflow baseline and paired measurement (Section 1)	D25

Gap	Correction designed in this edition	Validation
G09 - structural defects	Unique sections 0-22; one test/roadmap/demo location	D26

G01-G09 are specification fixes and acceptance contracts, not evidence that working source code has already been deployed.

3. Feature inventory: 64 explicitly buildable capabilities

Use this as the sprint backlog and regression audit. All 64 items have an ID, priority, implementation target and an observable acceptance demonstration. Parent categories are functional workstreams, not additional marketing features.

A / Institution intelligence

F01 Permanent-ID institution master [P0]

Build behavior: Authoritative ID, name, region, address and active history; PID never fuzzy-auto-linked.

Proof / test: Search exact PID; validate no cross-PID data mixing.

F02 Campus and institute hierarchy [P0]

Build behavior: Parent/campus/year validity with explicit uncertainty and reviewer approvals.

Proof / test: Wrong-campus names do not auto-collapse.

F03 Name/alias resolution [P0]

Build behavior: Verified aliases; fuzzy candidates only; provenance of every alias.

Proof / test: Find institute by alternate name.

F04 Institution 360 summary [P0]

Build behavior: Cases, source status, bureau ownership, action history, due work, location.

Proof / test: RB/PGRC entries shown with visibility boundaries.

F05 Institution-wide activity ribbon [P1]

Build behavior: Chronological action/status sequence across permitted sources.

Proof / test: View two separate cases without blending procedural status.

F06 Institution relationship graph [P1]

Build behavior: Typed case links with evidence; no inferred transitivity.

Proof / test: Select link and inspect reasons.

B / Intake and case management

F07 Multi-source complaint intake [P0]

Build behavior: Synthetic adapters and manual authorized case entry; unique case ID.

Proof / test: Create cases from two bureaus and refresh.

F08 Complete case metadata [P0]

Build behavior: Receipt/category/status/source/assignee/action dates/next action/due date.

Proof / test: All PS fields displayed and persisted.

F09 Native + analytical status mapping [P0]

Build behavior: Versioned native status and mapped status stored separately.

Proof / test: View native and harmonized views side-by-side.

F10 Editable officer work queue [P0]

Build behavior: Searchable role-scoped caseload, assign/reassign with recorded reason.

Proof / test: Officer assignment changes survive reload.

F11 Action timeline and notes [P0]

Build behavior: Append-only activity with actor, event/observed time and remarks.

Proof / test: Inspect sequence after action edit.

F12 Transfer acceptance and return [P0]

Build behavior: Recipient acknowledgement with pending/accepted/rejected and time.

Proof / test: Recipient rejects transfer and reason remains auditable.

F13 Evidence and supporting documents [P0]

Build behavior: Metadata, hash, secure scoped file reference, file access audit.

Proof / test: Unauthorized role cannot download attachment.

F14 Case relationship management [P1]

Build behavior: Link/keep separate/false match with exact identity of distinct cases.

Proof / test: Separate proceedings remain separate.

F15 Bulk import with preview [P1]

Build behavior: CSV template, validation, dry-run, row error report, idempotent commit.

Proof / test: Replay a batch with zero duplicates.

F16 Case lifecycle correction and reopen [P1]

Build behavior: Authorized state changes as new events; original state intact.

Proof / test: Reopen produces recorded reversal, not overwrite.

C / NEXUS SYNC flagship innovation

F17 Verified institution candidate blocking [P0]

Build behavior: Generate case pairs only within trusted identity/campus rules.

Proof / test: Different PIDs never auto-same institution.

F18 Explainable overlap review cards [P0]

Build behavior: Positive reasons, counter-signals, source dates, scope and uncertainty.

Proof / test: Different wording suggests relation; unrelated same PID stays distinct.

F19 Human reviewer dispositions [P0]

Build behavior: Coordinate/Keep Separate/False Match/More Info/Defer with mandatory reason.

Proof / test: Decision saved with actor and rule version.

F20 Blind cross-bureau handshake [P0]

Build behavior: Send opaque prospective overlap notice to authorized counterpart.

Proof / test: Other bureau receives no restricted narrative.

F21 Ack + lead/support assignment [P0]

Build behavior: Accept/decline with reason, nominate lead, create responsibility map.

Proof / test: Switch role and accept ping; lead persists.

F22 Coordination task board [P0]

Build behavior: Owner, description, dependencies, dueAt, state, evidence and acknowledgement.

Proof / test: Complete cross-bureau task end to end.

F23 Pre-action collision checker [P1]

Build behavior: Compare action class/purpose/doc type/time using deterministic rules.

Proof / test: Change proposed document class; warning updates.

F24 Action Passport [P1]

Build behavior: Display preflight finding, connected source coverage, permitted advisory outcome.

Proof / test: Shows CLEAR/REVIEW/UNKNOWN with reason.

F25 Scoped consent to reveal case context [P1]

Build behavior: Time-bound approved field disclosure with reason and audit.

Proof / test: Expired grant hides private text again.

F26 Coordination outcome receipt [P1]

Build behavior: Evidence snapshot, responsibilities, acknowledgements and outcome chronology.

Proof / test: Download role-redacted receipt matching events.

F27 Contradictory action monitor [P1]

Build behavior: Detect source-disagreeing statuses or incompatible action intents.

Proof / test: Record conflict without falsely attributing officer delay.

F28 Multilingual semantic reranker [P1]

Build behavior: Finals-target ranker, human-audited and measured; versioned rules remain fallback.

Proof / test: Show tested gain only with held-out data.

D / Dashboard, search and analytics

F29 Real-time/near-real-time command dashboard [P0]

Build behavior: Accurate totals and as-of timestamp; scoped updates or polling.

Proof / test: Counts change when a case is entered.

F30 All mandatory status counts [P0]

Build behavior: Pending, under process, disposed, transferred, escalated, overdue.

Proof / test: Dashboard totals reconcile with filtered list.

F31 Bureau and regional workload [P0]

Build behavior: Queues by bureau/region/officer with scoped drill-down.

Proof / test: Regional and bureau reports agree.

F32 Universal compound search [P0]

Build behavior: PID/name/case ID/bureau/region/category/status/date range.

Proof / test: Combine three filters and export same result.

F33 Cases-by-institution panel [P0]

Build behavior: Institution-wise case status and source-health context.

Proof / test: Filter institution and inspect every visible case.

F34 Scheduled reporting and CSV/PDF export [P0]

Build behavior: Scope-respecting, definition-stamped, audit-logged exports.

Proof / test: Download bureau/region/institute/status report.

F35 Pending ageing and disposal metrics [P0]

Build behavior: Bucket ages, aging trends, disposal denominator and date window explicit.

Proof / test: Ageing changes after clock advances in tests.

F36 Overlap effectiveness measures [P1]

Build behavior: Candidates, accepted links, dismissed links, coordination completions.

Proof / test: Never label alerts as avoided actions without evidence.

F37 Management executive summary [P1]

Build behavior: Task bottlenecks, stale feed gaps, overdue groups, relevant trends.

Proof / test: Inspect scope and counts underlying each card.

F38 Time-comparison reports [P1]

Build behavior: Month/quarter periods by reported definitions, not extrapolation.

Proof / test: Change dates and see recalculation.

E / SLA, alerts and follow-up

F39 Configurable SLA rule registry [P0]

Build behavior: Category, bureau, event, time zone, pause/holiday and approval version.

Proof / test: Rule edit creates version, not retroactive rewrite.

F40 Pending/near due/overdue alerts [P0]

Build behavior: Pre-due, breach, transfer waiting and assigned task reminders.

Proof / test: Alert triggers and is deduplicated.

F41 Controlled escalation workflow [P0]

Build behavior: Role-based escalation level and reason; management acknowledgment.

Proof / test: Escalate overdue case without modifying source record.

F42 Role-aware notification centre [P0]

Build behavior: Unread, acknowledge, dismiss, link to authorized record.

Proof / test: Notifications persist after refresh.

F43 Clock integrity and pauses [P1]

Build behavior: Distinguish official event time from sync-observed time; pause reasons.

Proof / test: Late data does not misclassify officer delay.

F44 Delivery outbox and retries [P1]

Build behavior: Deduplicated event delivery with at-least-once safe consumers.

Proof / test: Retry failure without duplicate notification.

F / Trust, privacy, governance

F45 Authentication and demo accounts [P0]

Build behavior: Secure sessions; RB/PGRC/management/reviewer seeded synthetic users.

Proof / test: Guest denied all protected routes.

F46 Server-side RBAC and field ABAC [P0]

Build behavior: Policy checked on read, write, search, export, socket and attachment.

Proof / test: Cross-role API returns no restricted fields.

F47 Full audit timeline [P0]

Build behavior: Log who did what, when, to which source, with reason and policy version.

Proof / test: Rejected access/override also recorded.

F48 Source provenance and freshness [P0]

Build behavior: Current/Stale/Conflicted/Not connected; source payload revisions and watermarks.

Proof / test: Unconnected source shows UNKNOWN not zero.

F49 Idempotent & conflict-preserving sync [P0]

Build behavior: Deduplicate per (system,ref); preserve source contradictions.

Proof / test: Reimport unchanged batch yields zero new cases.

F50 Accessible, responsive government UI [P0]

Build behavior: Keyboard, labels, contrast, focus, zoom, semantic status and printable view.

Proof / test: Core journey works at 200% zoom via keyboard.

F51 Audit integrity proofs [P1]

Build behavior: Hash chain with controlled storage and tamper-evidence validation.

Proof / test: Mutated event fails verifier without claiming blockchain.

F52 Data retention and purge policy hooks [P1]

Build behavior: Classification, retention labels, approval for deletion and archival.

Proof / test: Expired sample records eligible for approved policy.

F53 Import health and dead-letter inspection [P1]

Build behavior: Connector status, errors, watermark, quarantine and rerun job.

Proof / test: Malformed row is quarantined, not silently dropped.

F54 Field-level disclosure approval [P1]

Build behavior: Justification, issuer, permitted scope, expiry, reversal.

Proof / test: Disclosure expires without role escalation.

G / UX, quality, operations

F55 Stateful navigation and deep links [P0]

Build behavior: Stable routes for institution/case/task with permission checks.

Proof / test: Reload selected entity directly.

F56 Loading/empty/error/unknown states [P0]

Build behavior: Clear recovery messages, not zero or blank screens for missing feeds.

Proof / test: Source outage and partial loading tested.

F57 Synthetic dataset and reset utility [P0]

Build behavior: Reproducible fictional cases, overlap positives/hard negatives and demo users.

Proof / test: One-command reset produces same fixture hashes.

F58 Automated acceptance and security tests [P0]

Build behavior: API, unit, contract, end-to-end, negative authorization and replay cases.

Proof / test: Run published test command and inspect raw results.

F59 Guided in-app judge walkthrough [P1]

Build behavior: Role switch scenario, explanation overlay and reset.

Proof / test: Six-minute scripted walkthrough repeatable.

F60 Operational error monitor [P1]

Build behavior: Structured logs, health endpoint, request IDs, non-sensitive telemetry.

Proof / test: Failed connector diagnosed from correlation ID.

F61 Optional advanced trend explanations [P2]

Build behavior: Verified aggregate trend summarization with sources; no invented causes.

Proof / test: Summary grounded to report rows.

F62 Optional language translation assist [P2]

Build behavior: Reviewer-editable translated synopsis, source retained.

Proof / test: Original complaint always viewable with language tag.

F63 Optional external integrations [P2]

Build behavior: AICTE source APIs, official SSO, approved messaging only after agreements.

Proof / test: Connector never claims live use before authorization.

F64 Optional point-in-time audit replay [P2]

Build behavior: Replay versioned events and permissions at past cut-off; access audited.

Proof / test: Reconstruct older state deterministically.

4. Major modules: detailed implementation behavior

4.1 Institution 360

Officer-visible behavior. A canonical institution object is keyed only by authoritative verified Permanent ID. The display resolves approved name aliases and campus boundaries. The panel shows name, address, region, verified ID, case counts grouped by bureau and native/mapped status, linked case candidates, transfer state, the latest attributable action, next task, and source freshness.

Implementation. GET /institutions/:id aggregates only records the logged-in actor may see. Different source case identities remain separate; source feeds not connected produce UNKNOWN rather than an apparent zero.

Acceptance and edge cases. Case list filters and case URLs remain stable. Each displayed count declares its source scope and as-of timestamp.

4.2 Case management

Officer-visible behavior. Each incoming case has local UUID and an immutable (sourceSystem,sourceCaseRef). Store receipt, complaint text classification, category, native and mapped status, current bureau/officer, last event, next required action, due date, documents, and actor-attributed timeline.

Implementation. Never overwrite action history; mutation writes versioned case projection + append-only case_events in an atomic database transaction where required.

Acceptance and edge cases. Concurrent edits must reject stale entity versions and ask user to reload. Closing one coordinated task must not auto-close the source case.

4.3 Intelligent institution matching

Officer-visible behavior. First exact verified PID; then campus checks. String similarity, alias, regional and document clues produce an identity-review candidate only. Similar institution names with different verified PIDs are hard negative barriers.

Implementation. Normalize accents and punctuation for search but not for authoritative identity; each tentative identity assertion keeps a reviewer and proof source.

Acceptance and edge cases. No fuzzy-only automatic link. Where identity evidence is insufficient, mark unresolved and suppress unsafe aggregated conclusions.

4.4 Cross-bureau case relationship review

Officer-visible behavior. Within safe identity groups, produce candidate pairs from issue category, event timing, described incident, referenced documents, affected timeframe and requested remedy; list both supporting and contradicting factors.

Implementation. Assign relationship type (potentially related, same incident candidate, parallel proceeding, duplicate action candidate, keep separate, false match). Reviewer supplies disposition and reason.

Acceptance and edge cases. Same institution is not enough: unrelated cases and proceedings remain separate. Reviewers can reverse earlier decisions only through attributed correction records.

4.5 NEXUS SYNC handshake

Officer-visible behavior. A potential overlap creates a reviewer item. After review, a blind notice to another bureau contains opaque identifiers and non-sensitive coordination topic only. The recipient acknowledges or declines; a lead reviewer creates owned tasks and due dates.

Implementation. State machine permits controlled role-specific transitions and an explicit unresolved/conflict pathway. Notifications use an idempotent outbox. Access grants are independent of coordination status.

Acceptance and edge cases. Demonstrate RB->PGRC acknowledgement, named lead, completed task, and a receipt; denied RB access to private PGRC complaint content remains enforced.

4.6 Pre-action overlap checking

Officer-visible behavior. Before drafting an official document request, officer enters target institution, action type, document class, purpose and period. Engine examines known active action metadata and emits CLEAR, REVIEW or UNKNOWN with actual rule reasons.

Implementation. Read-only/advisory prototype. If source data is absent, UNKNOWN; no automated blocking or new statutory authority. Do not mislabel a draft comparison as a Digital Twin.

Acceptance and edge cases. Change proposed document class and show warning appearance/removal with fixture cases. Distinct proceedings or periods should not be blocked.

4.7 Transfer, ownership and escalation

Officer-visible behavior. Officer creates a transfer request with reason, target bureau, prior owner and relevant documents. Receiver can accept/reject; unresolved transfers have a pending status, reminder and full movement history.

Implementation. Lead officer and senior manager can escalate based on delegated scopes and policy version. Every task action requires actor, time and comment.

Acceptance and edge cases. No ownership is silently reassigned before recipient ack. Failed transfer and reminders remain visible to the original officer.

4.8 SLA and alert policy

Officer-visible behavior. Explicit policy records define target duration, clock start/end, business-day calendar, pause triggers, effective version and escalation contacts per relevant complaint category.

Implementation. Event time is separate from feed observation time. Due flags are calculated from approved policy + relevant task events; no fixed 21-day rule applies to all AICTE complaints by assumption.

Acceptance and edge cases. Timeline tested on boundary day, local timezone, pause, transfer and late import. Alert outbox retries without duplication.

4.9 Reports and analytics

Officer-visible behavior. Bureau-wise, region-wise, institution-wise and status-wise dashboards; total/pending/under-process/disposed/transferred/escalated; overdue; source coverage; average/median pending age; disposal rate with denominator.

Implementation. Use the same query builder for UI, CSV and PDF export. Each metric has definitionVersion, sourceScope, asOfTime and visibility restrictions. Show small group suppression.

Acceptance and edge cases. A clicked KPI must lead to a case table whose count reconciles. Never display missing-feed populations as measured zeros.

4.10 Data ingestion, coverage and provenance

Officer-visible behavior. Adapters accept source namespace, case key, authoritative field map, source revision, sourceUpdateAt, observedAt, access policy and batch ID. Store raw immutable snapshot and re-computable canonical projection.

Implementation. Replay is idempotent, conflicting records preserved with source claims and reconciliation owner, stale feeds flagged. Imported data is not evidence of official API approval.

Acceptance and edge cases. Corrupt CSV quarantined; unchanged replay makes zero additional case records; out-of-order status updates do not erase later official actions.

4.11 Secure evidence exchange

Officer-visible behavior. Default alert sees only opaque overlap and minimal shared references. Request access to restricted narrative/docs with reason, target reviewer, purpose, allowed fields and expiry.

Implementation. Backend enforces field rules in search, APIs, Socket.IO emissions, notification, exports and file downloads. Security event audit records reveals/denials/grants.

Acceptance and edge cases. Attempt GET/download/export after grant expiry and confirm sensitive content denied.

4.12 Administration and audit

Officer-visible behavior. Admin configures source registries, status mappings, calendars, category definitions, role grants, import health, controlled demo reset and audit search.

Implementation. Operational admin has no blanket access to all case narratives. Audit is append-only and can be hash-chained, but hash-chaining alone is not a certified tamper-proof ledger.

Acceptance and edge cases. Configuration changes versioned. Auditor views meaningful traces; destructive audit edit is rejected.

5. System architecture and deployment

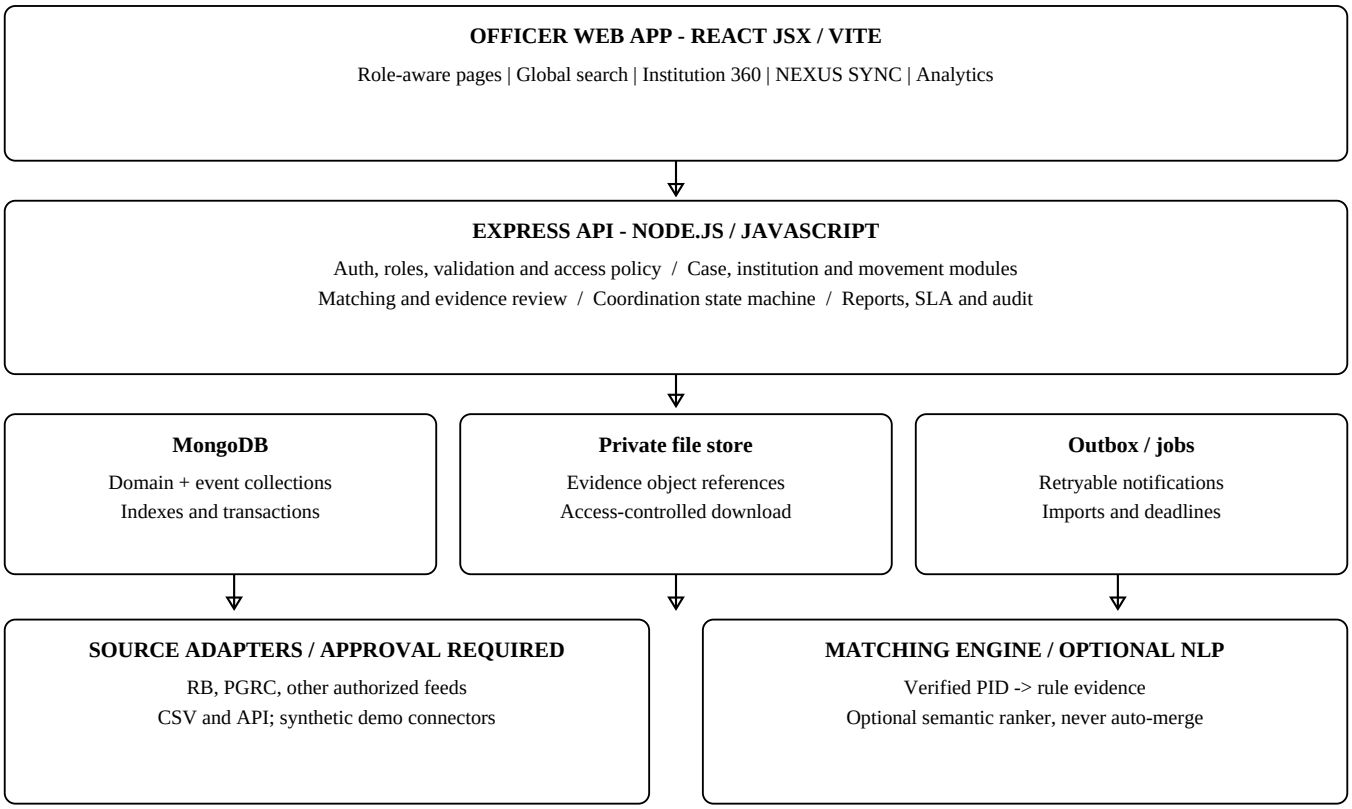


Figure: Five-tier modular JavaScript design. Authorized feed adapters are separate from the internal matching service; the optional NLP feature is not on the critical path.

Why one Node.js modular monolith

A modular Express application with domain services reduces same-night deployment complexity. Unit boundaries are institution, case, ingestion, matching, findings, coordination, action-preflight, alerts/SLA, reporting and admin/audit. Contract tests allow later service separation without building extra microservices prematurely.

Layer	Concrete choice	Contract
UI	React JSX, Vite, React Router, Tailwind, Recharts, TanStack Query	No .ts/.tsx; keyboard operation; role-aware page layouts.
API	Node.js LTS, Express in .js, Zod/Joi equivalent validators	REST, structured error codes, pagination and correlation ID.
Persistence	MongoDB + Mongoose, replica set for transaction-dependent writes	Compound unique keys; indexes; outbox and events.
Files	Private local object store for demo; S3/MinIO deploy option	Signed access or server proxy; never expose raw keys.
Async	In-process dev worker with durable MongoDB outbox; Redis/BullMQ optional	Retries/idempotence and scheduling must remain verifiable.
Matching	Verified PID + RapidFuzz-like JS similarity / deterministic evidence rules	Optional Python/embedding model only after measured validation.
Testing	Vitest, RTL, Supertest, Playwright, k6 optional	Prove E2E, role negative tests, performance metrics.

Deployment topology

1. Build React into static /dist and host behind a reverse proxy on HTTPS.

2. Run Express API as private application service; configure restrictive CORS, secure session cookies, rate limits and server-side role policy.

3. Run MongoDB (transactions require replica set); use non-production synthetic data for demo.

4. Store attachments privately; keep secrets outside repository; rotate demo credentials.

5. Run seed/reset and regression suites before jury demo; collect test reports, build hash and known limitations.

Project structure (.jsx + .js only)

```
nexus-ic/
  client/src/pages/*.jsx      # dashboard, institution, case, sync, reports
  client/src/components/*.jsx # cards, filters, tables, timeline, evidence
  client/src/services/api.js  # scoped server requests
  server/src/routes/*.js      # auth, cases, institutions, findings, sync
  server/src/domain/*.js      # ingestion, matching, policies, reporting
  server/src/models/*.js      # Mongoose schemas
  server/src/policies/*.js    # role + field checks
  server/src/jobs/*.js        # outbox, deadline, import retries
  tests/{unit,api,e2e}/*      # fixture-backed verification
  scripts/{seed,reset}.js     # synthetic reproducible scenario
```

6. Database design and transactional rules

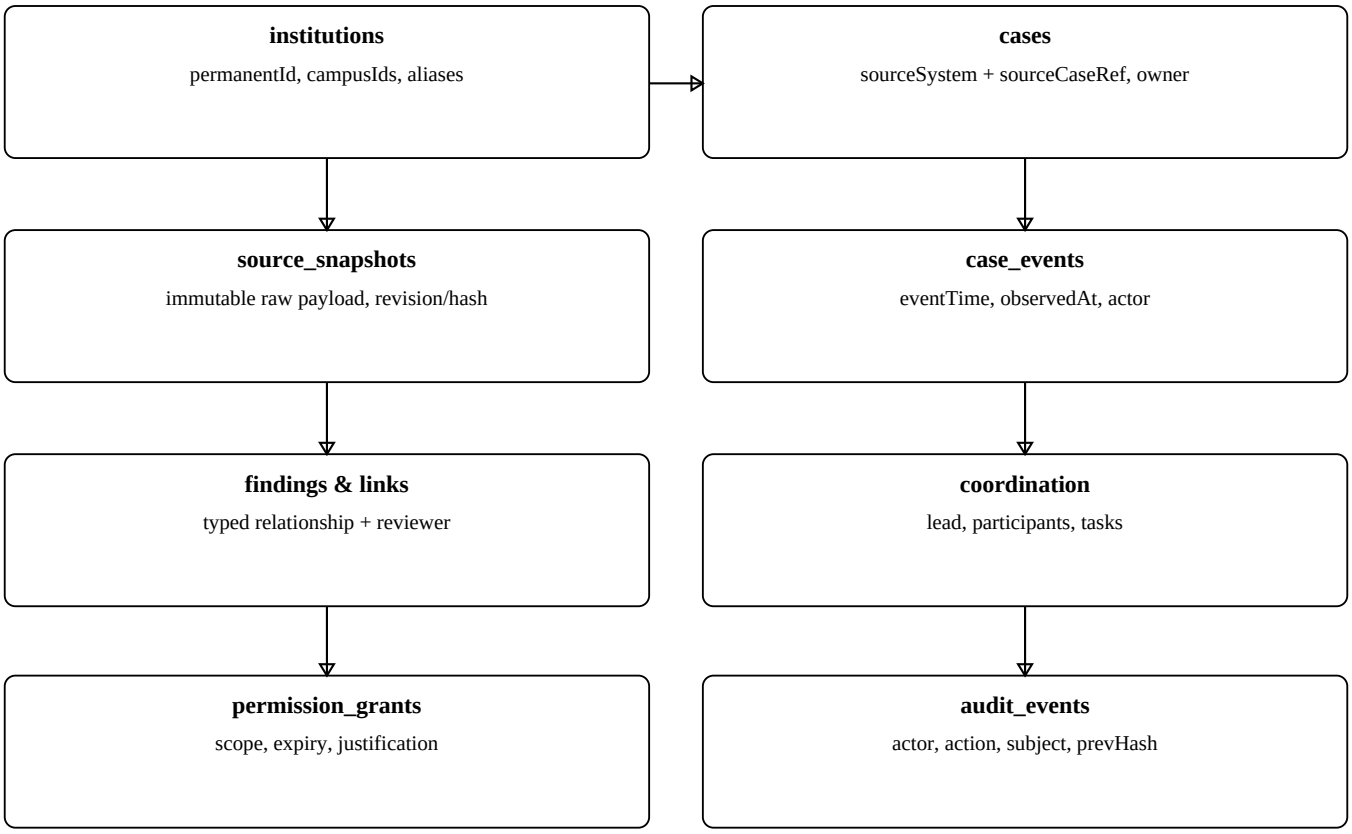


Figure: Document relationships: source snapshots and event chronology are preserved; no graph database is required.

Collection	Key fields	Integrity rule/index
institutions	permanentId(unique), name, region, address, activeFrom, metadata	Permanent ID unique; region/campus indexes

Collection	Key fields	Integrity rule/index
campuses	institutionId, campusRef, name, validYears	(institutionId,campusRef) unique
institution_aliases	institutionId, alias, proofSource, reviewedAt	alias text + institutionId
identity_assertions	candidateA/B, evidence, reviewer, disposition, ruleVersion	no automatic fuzzy binding
source_systems	sourceCode, owner, authority, supportedFields, heartbeat	unique sourceCode
sync_runs	sourceCode, batchId, watermark, counts, errors, startedAt	(sourceCode,batchId) unique
source_snapshots	sourceSystem, sourceRef, revision, rawHash, rawPayload, observedAt	immutable and unique source revision
cases	caseId, sourceSystem, sourceRef, institutionId, bureau, officer, nativeStatus, mappedStatus, category, dueAt, version	unique (sourceSystem,sourceRef); indexes for search
case_events	caseId, eventType, actor, eventTime, observedAt, remarks, prevVersion	append-only by caseId/eventTime
case_movements	caseId, fromBureau, toBureau, requestedAt, ackAt, response	index (toBureau,state)
evidence_refs	caseId, storageKey, mime, fileHash, classification, accessPolicy	no raw private document in case list
case_relationships	caseA, caseB, typedRelation, positive, negative, reviewReason	unique ordered pair + relation type; no transitivity
match_findings	caseA, caseB, kind, modelVersion, ruleVersion, priority, state	index state and createdAt
finding_dispositions	findingId, reviewerId, decision, reason, at	append-only
coordination_records	leadBureau, participants, findingIds, state, disclosureScope	index lead + state
coordination_tasks	coordinationId, ownerId, dependencies, dueAt, ackAt, evidence	index owner + dueAt
action_proposals	actor, institutionId, purpose, documentClass, timePeriod, preflight	idempotencyKey unique
notifications	recipientId, kind, opaqueSummary, deliveredAt, ackAt	recipient+dedupeKey unique
sla_policies	policyKey, scope, businessCalendar, period, pauses, validFrom, version	immutable rule versions
access_grants	subject, scope, purpose, approver, expiry, revocation	index grantee and expiry
rule_versions	mapping, reviewerThresholds, policyApproval, effectiveAt	version immutable
audit_events	actor, resource, action, reason, occurredAt, prevHash, hash	append-only; retention governance

Critical invariants

1. Source-native case ID is never overwritten by local case ID. Unique (sourceSystem,sourceCaseRef) prevents duplicate ingest.

2. Every source snapshot is immutable; logical corrections create new revisions or attributed events.

3. Case relationships are typed pairwise and not transitively inferred; no automatic case merge/closure.

4. Version checks are required on concurrent officer and coordination mutations; idempotency keys for retries.

5. Audit events are append-only and authorized evidence/download events include actor, subject, purpose and policy decision.

6. Views/exports must use the same backend policy; source gaps yield UNKNOWN.

Representative case shape - abbreviated

```
{
  caseId: "C-00031", sourceSystem: "PGRC-DEMO", sourceCaseRef: "PG-31",
  institutionId: "TEST-PID-001", category: "Fee/Certificates",
  nativeStatus: "Under Scrutiny", mappedStatus: "UNDER_PROCESS",
  currentBureau: "PGRC", assignedOfficerId: "demo-pgrc-01",
  receivedAt: "2026-10-19T10:30:00+05:30",
```

```
  lastActionAt: "...", nextAction: "Await documents", expectedDisposalAt: null,
  policyVersion: "DEMO-1", sourceRevision: 2, version: 5,
  classification: "RESTRICTED", sourceScope: "PGRC-DEMO"
}
```

7. API endpoint specifications and response discipline

The table defines a proposed application API, not existing AICTE endpoints. Every API is server-authorized, validated, paginated where relevant, with standardized errors: VALIDATION_ERROR, NOT_AUTHORIZED, NOT_FOUND, STALE_VERSION, FEED_UNKNOWN, RULE_NOT_CONFIGURED and INTERNAL_ERROR.

Verb	Endpoint	Purpose	Mandatory guarantee
GET	/api/v1/me	Return role and authorized scopes	role claim verified server-side
GET	/api/v1/institutions	Search PID/name/region/campus	search filters by authorized scope
GET	/api/v1/institutions/:id	Institution 360 projection	response redacts hidden bureau fields
GET	/api/v1/cases	Filter cases and paginate	parameter whitelist, allowed scopes
POST	/api/v1/cases	Authorized manual case creation	unique source key, input validation
GET	/api/v1/cases/:id	Case detail + native and mapped state	source policy + field-level filters
POST	/api/v1/cases/:id/events	Append case action or correction	optimistic version, audit event
POST	/api/v1/cases/:id/transfers	Request movement	recipient ack required
POST	/api/v1/transfers/:id/respond	Accept/reject transfer	only target officers
POST	/api/v1/imports/preview	Validate CSV feed without commit	quarantine invalid rows
POST	/api/v1/imports/commit	Idempotent import batch	unique(sourceSystem,sourceRef)
GET	/api/v1/findings	List scoped overlap candidates	opaque across restricted bureaus
POST	/api/v1/findings/:id/disposition	Coordinate/Keep Separate/False Match	mandatory reason and role
POST	/api/v1/coordination/requests	Send blind handshake	opaque payload and audience guard
POST	/api/v1/coordination/:id/respond	Accept/decline/request info	version check, authorized participant
POST	/api/v1/coordination/:id/tasks	Assign accountable task	lead/delegate authorization
PATCH	/api/v1/tasks/:id	Update task state with evidence	audit status transition
POST	/api/v1/actions/preflight	Advisory action collision result	CLEAR/REVIEW/UNKNOWN not legal ruling
POST	/api/v1/access-requests	Request time-limited reveal	approval path, default deny
GET	/api/v1/reports	KPI data with scope/as-of metadata	privacy suppression
GET	/api/v1/reports/export	Audit-logged CSV/PDF	same authorization as screen
GET	/api/v1/notifications	Role inbox	no forbidden content in preview
POST	/api/v1/notifications/:id/ack	Ack alert	idempotent
GET	/api/v1/data-health	Connector status + conflicts	admin or delegated reader
GET	/api/v1/audit/:resourceId	Authorized trace of events	append-only log filtering
GET	/api/v1/health	Liveness/readiness	non-sensitive public response

Example - candidate review and coordination

```
POST /api/v1/findings/FND-004/disposition
{ "decision":"COORDINATE", "reason":"Same verified PID and request period",
  "expectedVersion":2 }
-> 200 { "findingId":"FND-004", "state":"COORDINATION_REQUESTED",
       "auditEventId":"EV-239", "caseIds":["RB-008","PG-031"] }
POST /api/v1/coordination/requests
{ "findingId":"FND-004", "targetBureau":"PGRC", "messageClass":"OVERLAP" }
-> 201 { "coordinationId":"CO-107", "recipient":"PGRC",
        "exposedFields":["institutionOpaqueId","caseRefOpaque"] }
```

Common API rules

- Responses identify asOfTime and scope when showing counts, source-cached state or analytics.
- Prevent IDOR: caller-controlled institutionId, officerId or sourceSystem is always verified against permitted scope.
- The server returns redacted objects, not frontend-hidden fields. Errors never return private details.
- Replay/duplicate writes respond idempotently and never create second tasks; optimistic version gates edits.

Additional APIs required by the nine corrections

Method	API route	Responsibilities
GET	/api/v1/authority/overview	Role-filtered KPIs, escalations, policy and approval inbox
POST	/api/v1/authority/decisions	Delegated sign-off + actor != requester and version checks
GET	/api/v1/institutions/:id/catalog	One-window catalog projection with per-case metadata redaction
GET	/api/v1/matching/explain/:pairId	Detailed contributions, raw permitted evidence and counter signals
GET	/api/v1/sla/policies	List demo vs approved policy versions, calendar and active state
POST	/api/v1/sla/policies/:id/versions	Create nonretroactive policy version; authority approval
POST	/api/v1/notifications/test	Admin/test-only email/sms simulator, no actual private data
POST	/api/v1/notifications/:id/ack	Idempotent acknowledgement and trace
GET	/api/v1/notifications/delivery-status	Retry/failed/dead letter audit for delegated admins

8. React JSX UI structure and state contracts

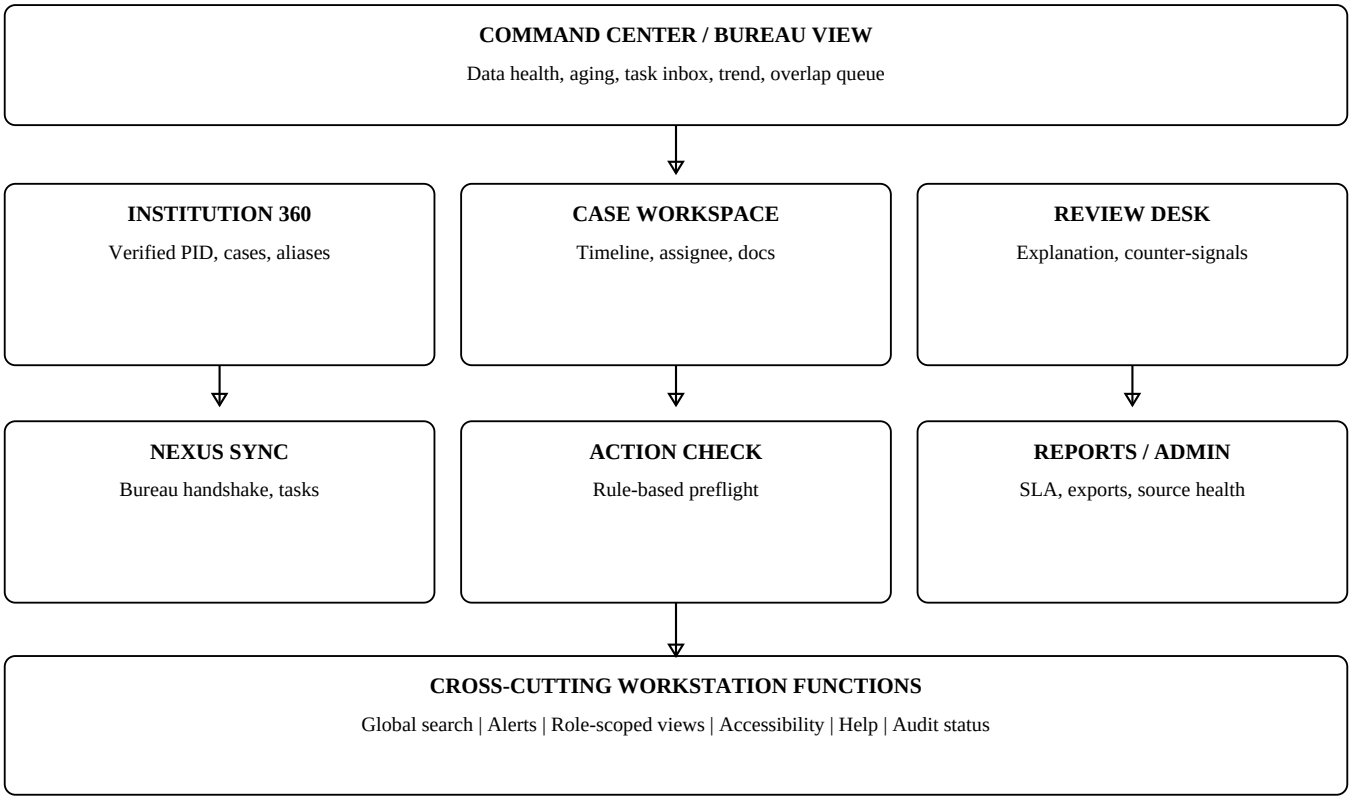


Figure: Navigation from command dashboard into institution, case, evidence review and coordination; cross-cutting search and accessibility.

Route / component	Major fields and interactions	Empty/error/permission state
/login.jsx	Demo role sign-in, invalid credentials, session expiry	No access to application state without server session
/dashboard.jsx	Status cards, aging, workload, urgent tasks, feed health	Unknown feed banner; never fake totals
/institutions.jsx	Search PID/name/region; aliases, campus markers	Ambiguous campus identity requires review
/institutions/:id.jsx	Institution 360, status, source coverage, activity ribbon	Private bureau record masked if not entitled
/cases.jsx	Filterable and exportable case list	No results vs no feed vs denied are distinct
/cases/:id.jsx	Native/mapped status, assignment, documents, complete timeline	Case edits lock version; audit visible
/findings.jsx	Evidence cards, false-match review and counter signals	Uncertain model gives review not decision
/sync.jsx	Blind requests, lead and tasks, acknowledgement, closure	No private content before scoped grant
/action-check.jsx	Proposed class/period/purpose and CLEAR/REVIEW/UNKNOWN	UNKNOWN for absent source
/reports.jsx	Ageing, bureau/region/institution/status, trends/export	Metric definition/time/coverage shown
/data-health.jsx	Connector sync errors, stale/conflict queue	Missing connector never displayed as healthy
/admin.jsx	Role policy, SLA registry, mapping, audit search	Only delegated administration routes

Important small UX behaviors - must not be missed

• Top bar: current role, permission scope, accessible global search, as-of time, demo banner, help and log out.

• List tables: sort, multi-filter, clear filters, result count, pagination, sticky header, keyboard focus, empty/loading/error states.

• Case views: copy safe reference, last/next action, status badge with text label, timestamp and timezone, source-native history, attachment metadata.

• Overlap cards: show both case refs, same-institute proof, event period, positive and counter evidence, uncertainty, source freshness and reason-required decision.

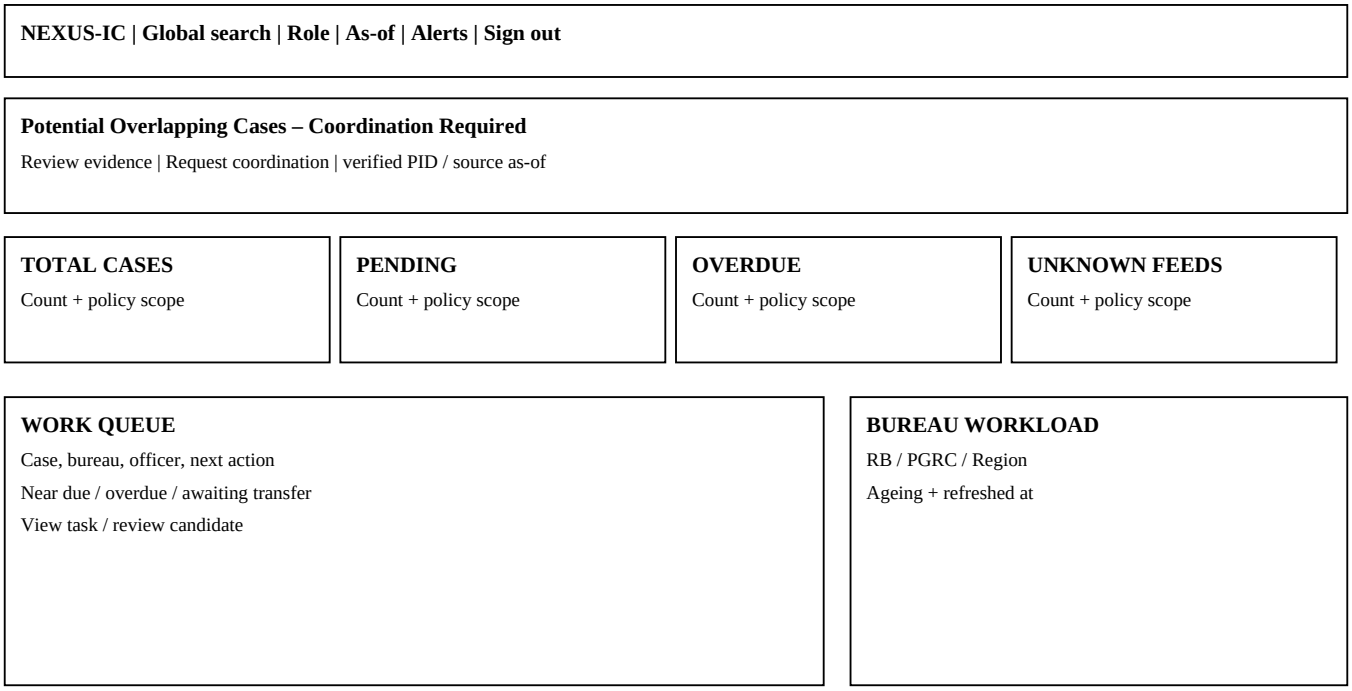
• Coordination panel: unread notice, approve/request info/decline, lead selection, owner task/due date, notes, ack/receipt, controlled close.

• Reports: date range, region/bureau/category/status selectors, provenance footnote, CSV export, print view, suppressed small populations.

• Mobile/tablet/desktop: no horizontal body overflow, 200% zoom, logical tab order, text alternatives and predictable dialog behavior.

Six complete screen wireframes - implementation reference

These are functional layout plans, not screenshots of an implemented React application. All six views must render loading, empty, denied, error and source-UNKNOWN conditions.



Wireframe: proposed layout; not a screenshot of deployed functionality.

Figure: Command Dashboard: KPIs, data coverage, exact overlap banner and due-work lists.

NEXUS-IC | Global search | Role | As-of | Alerts | Sign out

INSTITUTION 360 / TEST-PID-001

Name | Region | Campus / aliases | Verification state

SOURCE / ACCESS

RB current | PGRC current

Restricted details explicitly masked

STATUS SUMMARY

Cases by owner, mapped/native state

Overlaps / handovers / due alerts

UNIFIED CASE INDEX

RB-008 | RB | records | IN_PROCESS

PG-031 | PGRC | fee | IN_PROCESS

Selected confidential fields: RESTRICTED

View permitted details / request access

ACTIVITY RIBBON

Event -> bureau -> time

Transfer / source status

Evidence and as-of

Wireframe: proposed layout; not a screenshot of deployed functionality.

Figure: Institution 360: verifiable ID, unified case index, source-native statuses and timeline.

NEXUS-IC | Global search | Role | As-of | Alerts | Sign out

Potential Overlapping Cases – Coordination Required

FND-004 / RB-008 and PG-031 / source coverage

MATCHED CASES + POSITIVE

Same verified PID

Certificates / payment / period

Example score 93.75; demo only

COUNTER-SIGNALS / UNKNOWN

Distinct proceedings? campus?

Source stale? missing complaint period?

Evidence refs + explanation

DISPOSITION (REASON REQUIRED)

[Coordinate] [Keep separate] [False match] [Need information]

Reviewer, source version, decision time, rationale

REVIEWER TRACE

Prior decision / version / no automatic merge or case closure

Wireframe: proposed layout; not a screenshot of deployed functionality.

Figure: Overlap Review: evidence/counter-evidence, thresholds and human disposition.

NEXUS-IC | Global search | Role | As-of | Alerts | Sign out

PROPOSED ACTION

PID | type: document request

Purpose / doc class / period

Officer & source case / submit preflight

PREFLIGHT RESULT: REVIEW

Potential duplicate request

Known related action RB/PGRC

Connected feed / unknown scope

EVIDENCE & COUNTER-EVIDENCE

Same class / period / verified PID / related incident

Counter: separate proceedings, older dates, uncertain feed

Rule version / as-of; no autonomous hard block

OFFICER OPTIONS

[Proceed with recorded reason] [Edit action] [Start NEXUS SYNC]

Log decision and permitted action; no official action auto-executed

Wireframe: proposed layout; not a screenshot of deployed functionality.

Figure: NEXUS PREVENT: action proposal, rule result and approved alternatives.

NEXUS-IC | Global search | Role | As-of | Alerts | Sign out

NEXUS SYNC / CO-107

Blind notice | PGRC: accept / decline / ask information

Explicitly NO complainant narrative before approval

LEAD AND PARTICIPANTS

Lead Bureau: PGRC [Approve]

RB support / authority contact

Acknowledgements and timestamp

COORDINATION TASK BOARD

Collect prior documents | owner

Due time, state, dependencies

Escalate if unacknowledged

DECISION AND AUDIT RECEIPT

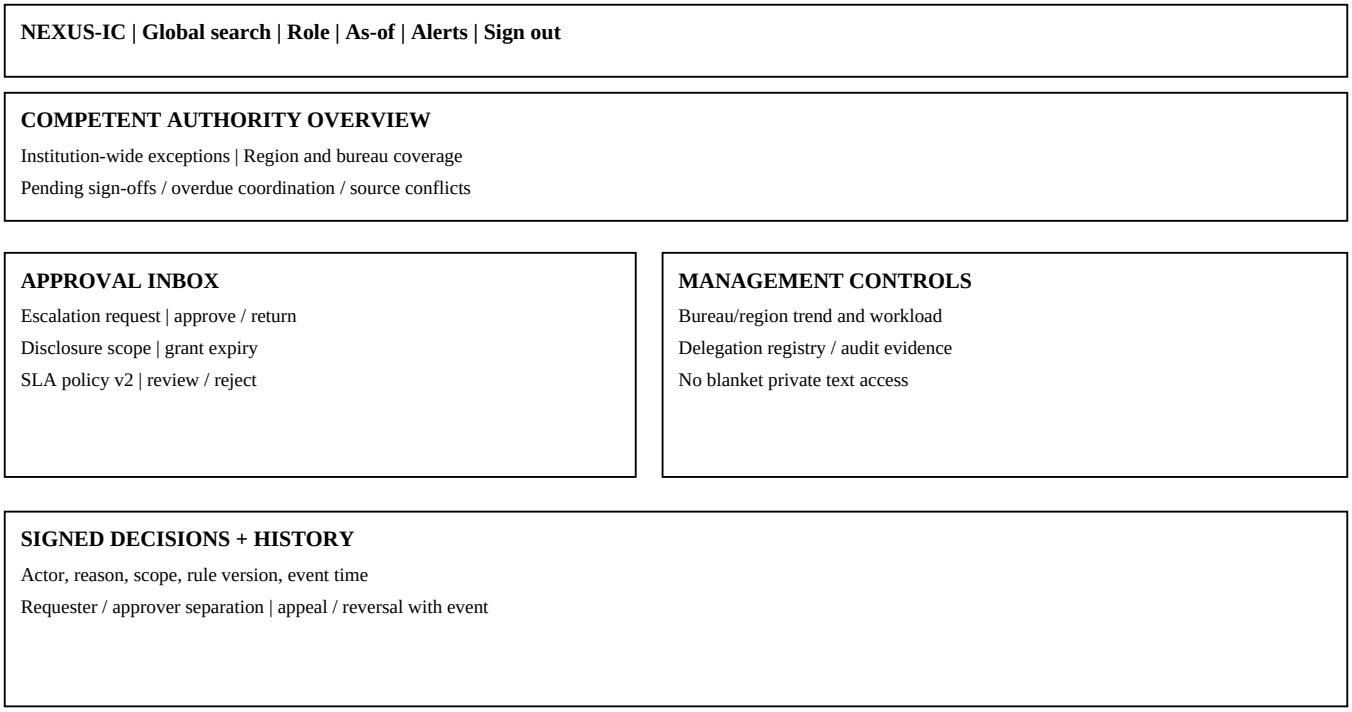
Typed relationship | exact source case IDs kept separate

Task evidence / reviewer sign-off / completion receipt

No auto closure; scope-limited download

Wireframe: proposed layout; not a screenshot of deployed functionality.

Figure: NEXUS SYNC: blind notice, bureau acknowledgement and owned tasks.



Wireframe: proposed layout; not a screenshot of deployed functionality.

Figure: Competent Authority: escalations, access policy, SLA approval and audit.

Required banner behavior - exact problem-statement text

Potential Overlapping Cases – Coordination Required	Display this exact wording (including capitalization and dash) as a text-labelled, keyboard-accessible alert on /dashboard, /institutions/:id and /review whenever an authorized pending candidate meets the configured review threshold. Include count, involved case references as permitted, as-of time, source-coverage note and action: Review evidence / Request coordination. Do not show it for confirmed false matches or merely same-institution unrelated cases.

9. Matching and advisory collision logic

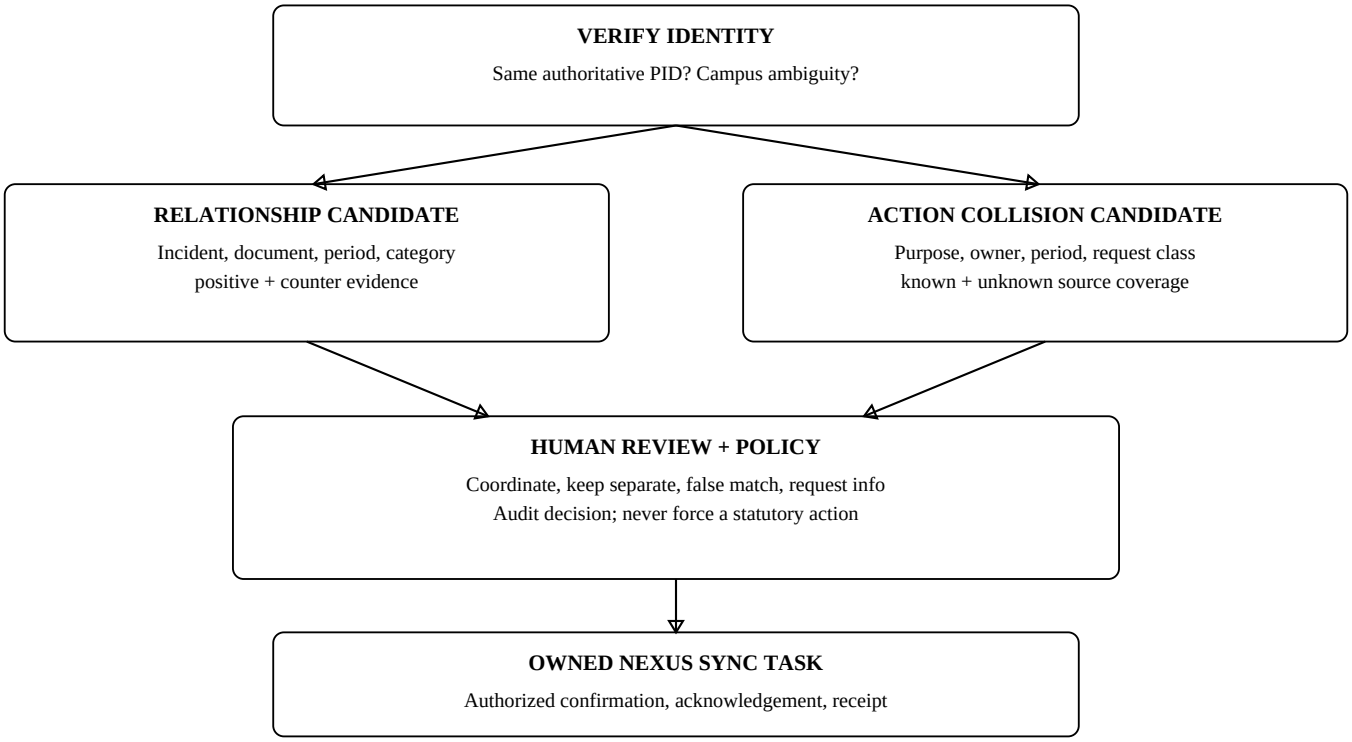


Figure: Three distinct operations: authoritative identity, suggestive incident relationship, and pre-action collision; human review is the only bridge to coordination decisions.

Identity and case matching pipeline

1. Candidate generation: restrict to verified Permanent ID or identity-review queue. Never equate similar names with the same institution.

2. Relationship features: source category, incident period, optional document reference overlap, issue and remedy terms, relevant bureau pair and counter-signals.

3. Create evidence card: exact source facts, last sync, positive/counter reasons, incomplete feed flags, versioned rule ID.

4. Review: human chooses Coordinate/Keep Separate/False Match/Request Information/Defer and writes the mandatory reason.

5. Only authorized accepted links enter NEXUS SYNC; no automatic merge or statutory decision.

Example deterministic collision rules (illustrative, not AICTE regulation)

Rule	Trigger	Result
C1	Exact verified PID + same doc class + same period + active action purpose	REVIEW: likely repeated document demand
C2	Different verified PID	CLEAR on identity; no cross-PID merge
C3	Same PID, different incident and non-overlapping period	No collision from identity alone
C4	PGRC feed disconnected/stale at point of decision	UNKNOWN with missing-coverage reason
C5	Different legal proceeding with same evidence request	REVIEW coordination only, never procedural merger

Model release validation, semantic integration and rule fallback

Only when labeled, approval-cleared data exists: split by institution (not random case pair) to avoid leakage; compare PID-only and rule-only baselines; report precision/recall/F1/PR-AUC and false-match review cost; calibrate thresholds. Embeddings may rank human-review candidates, but model text must not fabricate reasons. If the ML component fails, rules still run. Do not claim accuracy until reproduced.

Executable matching specification M-1 - not just a feature title

Stage A (institution identity): normalize Permanent ID and match exact institution plus validity period/campus. Different verified IDs are an authoritative NOT-SAME gate. Missing/ambiguous IDs go to a human identity-resolution queue, never automatic same-institution linking. Names alone create identity-review suggestions only.

Stage B (issue relationship): Within verified identity, compare two complaint incidents using seven independently inspectable signals. For each signal s in [0,1], the default demo score is:

Weighted evidence score	S = 25(issueConcept) + 15(timeWindow) + 10(category) + 15(documentOrSubject) + 10(remedyIntent) + 10(sharedEntitiesOrReferences) + 15(semanticSimilarity). Each term is weight times a normalized 0-to-1 evidence value. Weights sum to 100. Missing evidence scores zero and adds an explicit missing-evidence warning; an NLP outage does not silently inflate the score. Output is a relationship-review score, NOT probability of guilt or factual certainty.

Signal	Demo weight	How it is computed and evidenced
Issue concept overlap	25	Curated, versioned issue taxonomy and synonyms; e.g., retained certificates == withheld originals.
Incident-date proximity	15	1 for same/adjacent week, 0.5 for next 30 days, 0 if absent/far; adjustable by category.
Complaint category	10	1 same normalized category, .5 related family, 0 unrelated.
Document / subject	15	Exact normalized object class (original certificates, fee receipt, building inspection).
Requested remedy	10	Shared practical objective such as release originals or stop a duplicate document demand.
Shared entities / refs	10	Signed document ref, incident ID, names or scoped institution facts with confidence.
Semantic text similarity	15	Versioned multilingual sentence embedding normalized/calibrated from held-out reviewer labels.

Thresholds and safe disposition - demo starting values only

Score / condition	Proposed system behavior	Follow-up
Verified IDs differ	IDENTITY BLOCK: not same institution	No overlap alert; investigate bad source data only
80-100 and >=3 independent non-text signals	HIGH REVIEW candidate; prominent warning	Officer still decides; never auto-merge
60-79	REVIEW candidate; exact coordination-required banner	Evidence card, acknowledge and human disposition
40-59	LOW candidate in reviewer queue; no cross-bureau broadcast	Request documents or re-score after new data
0-39	No overlap banner by default	Record retrieval only, unless reviewer manually links
Unknown ID or stale source	IDENTITY REVIEW / COVERAGE UNKNOWN	Do not report CLEAR based on missing data
Explicit different incident or proceeding	Typed RELATED-BUT-SEPARATE / KEEP_SEPARATE	Counter-evidence prevents duplicate-action conclusion

Worked positive example - differently worded fictional complaints

RB-008 (RB)	PID TEST-PID-001; 10 Oct 2026; category Fee/Documents. Text: "The college retained original study certificates after our fee balance was cleared." Proposed remedy: return originals.

PG-031 (PGRC)	PID TEST-PID-001; 11 Oct 2026; category Student Records. Text: "Despite settlement of outstanding charges, the registrar will not hand back academic originals." Requested remedy: release papers.

Signal	Normalized evidence	Contribution
Issue concept	1.00 (same withholding issue)	25.00
Date window	1.00 (adjacent day)	15.00
Category	1.00 (same mapped family)	10.00
Subject class	1.00 (original certificates)	15.00
Remedy	1.00 (return originals)	10.00
Shared refs/entities	0.60 (institution + registrar context)	6.00
Semantic	0.85 (example model score, not a trained measurement)	12.75
TOTAL	Illustrative score: 93.75 / 100	93.75

Result: HIGH REVIEW; display "Potential Overlapping Cases – Coordination Required" with source case refs and counter-signals. The 0.85 semantic value is deliberately illustrative; do not claim a trained model produced it. Officers may still choose KEEP_SEPARATE.

Worked hard negative and failure fallback

Same PID, RB-009 complaint about laboratory fire-door maintenance (June), PG-031 about return of certificates (October). Example feature values issue=0, period=0, category=0, subject=0, remedy=0, shared=0.3 and semantic=0.1 produce S=4.5/100 (3 + 1.5); no banner. Same text across two different verified PIDs is identity-blocked regardless of textual score. If embeddings are unavailable, present RULE-ONLY ranking with explicit unavailability, and evaluate separate calibrated thresholds; never claim semantic capability is working when the service is off.

Calibration and implementation acceptance

• Build a minimum locked validation set: 40 true relation pairs plus 40 hard negatives, including at least 10 semantically paraphrased, 10 multilingual or translated and 10 identical-institution unrelated pairs; do not train and test on the same institution.

• Compute confusion matrix, precision, recall, F1, review queue precision, per-language results and false-positive costs. Freeze threshold/rule/model versions before demo and show errors.

• Semantic feature (F28) is a required finals-target integration in this final specification; if its evaluated result is worse than rules, disable by feature flag and transparently report rule-only fallback.

• Every evidence card shows contributions, source references, timestamps, counter-signals, provenance, candidate generation method and role-restricted fields. Reviewer decisions always require reasons.

10. NEXUS SYNC lifecycle, business controls and permissions

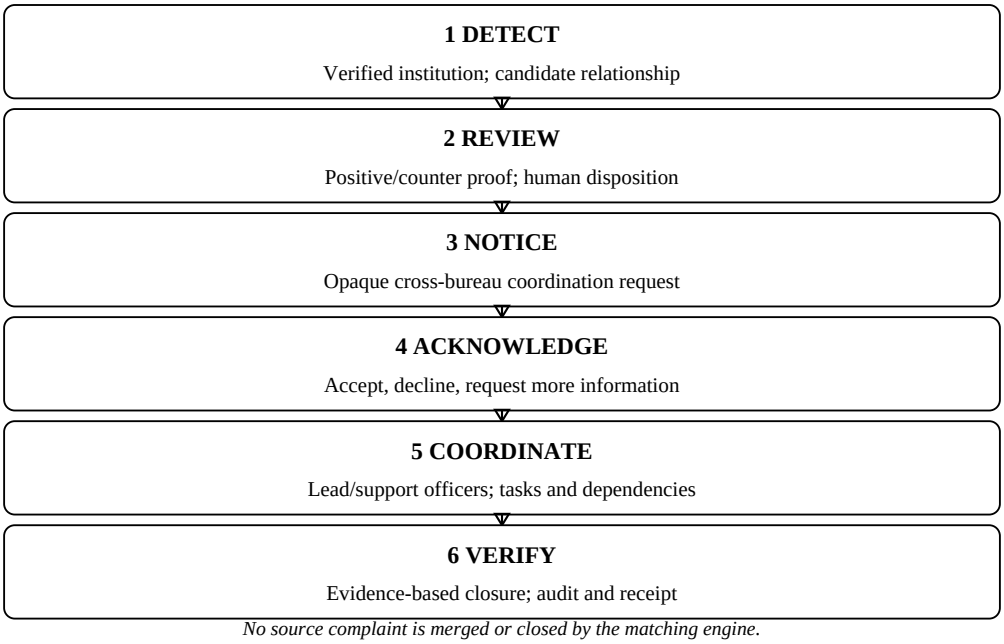


Figure: Default process: detect, review, notice, acknowledge, coordinate, verify. Keep Separate and False Match are alternative audited dispositions.

Coordination state machine

Current	Authorized event	Next	Mandatory side effect
CANDIDATE	Reviewer opens evidence	UNDER_REVIEW	Viewer and evidence version recorded
UNDER_REVIEW	COORDINATE decision	NOTICE_PENDING	Reason and typed case link recorded
UNDER_REVIEW	KEEP_SEPARATE / FALSE_MATCH	CLOSED_WITH_REASON	No coordinated task; corrections auditable
NOTICE_PENDING	Create blind notice	AWAITING_RESPONSE	Opaque event to authorized recipient
AWAITING_RESPONSE	Accept / decline / ask info	ACCEPTED / DECLINED / NEED_INFO	Responder, timestamp, reason captured
ACCEPTED	Assign lead, roles and tasks	IN_PROGRESS	Task/owner/due date + scoped visibility
IN_PROGRESS	All work acknowledged and signed	VERIFIED_CLOSED	Evidence and signed receipt created
ANY OPEN	Approved escalation or conflict	ESCALATED	Owner and escalation reason logged

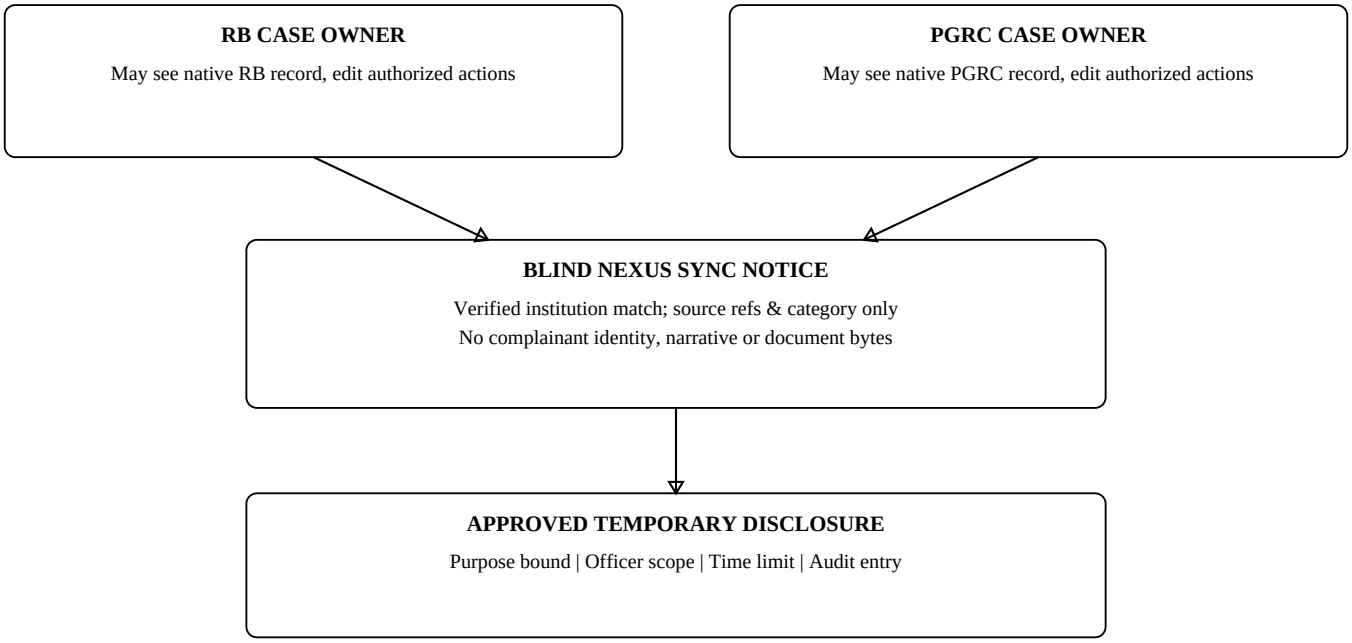


Figure: The coordination handshake does not itself grant access to restricted complainant identities, case narratives or documents.

Role and action matrix - proposed, approval required

Role	View policy	Action policy	Export scope
RB Officer	Unified cross-bureau safe-metadata catalog; own-bureau full details; other narratives masked	Accept/request coordination; own actions	Own assignments; no private cross-bureau export
PGRC Officer	Unified cross-bureau safe-metadata catalog; own-bureau full details; other narratives masked	Accept/request coordination; own actions	Own assignments; no private cross-bureau export
Regional Officer	Authorized region and approved case categories	Triage/reassign in delegated regional scope	Region reports with suppression
Reviewer / Coordinator	Case evidence when role+purpose+scope grants permit	Disposition candidate; lead assignment; coordination tasks	Authorized receipt only
Bureau Lead	Bureau-wide permitted views	Approve transfer/escalation as delegated	Bureau metrics, audit scoped
Senior Management	Scoped aggregates, exception catalog plus approved case detail	Approve escalations/policies if delegated	Management reports; limited PII
Competent Authority	Dedicated oversight dashboard, scoped exceptions and approvals; no automatic PII reveal	Delegated escalation, disclosure and policy approval with independent requester	Scoped management reports, attributable signed decisions
System Admin	Connector/identity/policy config; no blanket complaint narrative	Configure demo data, access grants subject to approval	Audit admin operations, not content by default
Independent Auditor	Read-only audit subset and permitted evidence	No operational decisions	Scoped compliance extracts

Access escalation principle

Default-deny disclosure: cross-bureau counterpart receives a minimum-data blind notice. If case text or documents are necessary, request a purpose-bound temporary grant approved by designated roles. Log requester, approver, legal basis/policy, specific fields, expiry, revocation and every view/export. Do not assume bureau-to-bureau narrative visibility is legally authorized.

Single-window contract: discoverability is not universal disclosure

An authenticated officer with a legitimate case-related purpose can search ONE institution-wide case catalog across connected and authorized bureau sources, even when a detailed narrative is restricted. Search results show only a minimum common schema: opaque case reference, verified Permanent ID, source bureau, coarse issue category, mapped status, assigned-office label, last action timestamp, source-as-of and existence of a potential coordination flag. Hidden cases may require even the metadata to be masked by category-specific policy; disclose the omission as "restricted entries" when permitted, never silently pretend completeness.

Viewer	Unified catalog	Case narrative/evidence	Permitted action
RB / PGRC owner	All permitted institution index entries; own bureau details	Own cases; other case text masked unless granted	Request coordination, propose action, modify own case
Regional officer	Regional case catalog with restricted-case placeholders	Region-authorized details only	Triage, transfer and escalate if delegated
Cross-bureau reviewer	Scoped catalog plus evidence-card allowed signals	Purpose-bound reviewed grant required for PII	Human overlap disposition, coordinated tasks
Senior Management	Institution/bureau aggregates and exception summaries	Case text only by separately approved delegation	Escalate and supervise through documented scope
Competent Authority	Dedicated institution-level control panel and policy/audit index	Only expressly authorized case detail; not automatic PII access	Approve delegated escalation, policy and disclosure; issue signed administrative oversight decisions
System administrator	Connector, user, role and import metadata	No automatic complaint narrative access	Configure permissions; cannot self-approve disclosure

Competent Authority dedicated screen and powers - delegated, not assumed

• Route /authority.jsx: total/pending/overdue cases with source coverage, bureau-wide handover failures, unresolved source conflicts, policy violations and pending approvals.

• Decision inbox: cross-bureau escalation requests, stalled transfer acknowledgements, scope-limited disclosure requests, approved SLA policy versions and assigned review delegations.

• Two-step sensitive approvals (requester != approver), required justification, affected field list, effective dates, expiry, electronic attribution and append-only audit.

• Authority does not overwrite native regulatory decisions or acquire unrestricted complainant identity merely because of position. Operating permissions require AICTE validation.

Single-window authorization invariants

• Server-side ABAC applies consistently to list counts, search, institution profile, knowledge graph, export, API, file downloads, notifications and sockets.

• Every catalog count shows scope, connected/unknown source coverage and asOfTime. Masked entries never leak via filters, facet counts, logs or export.

• A blind coordination notice does not automatically authorize access to sensitive complainant narrative. Temporary grants are field-, purpose-, actor- and time-bound, with expiry enforced server side.

11. Source integrity, timing and reconciliation

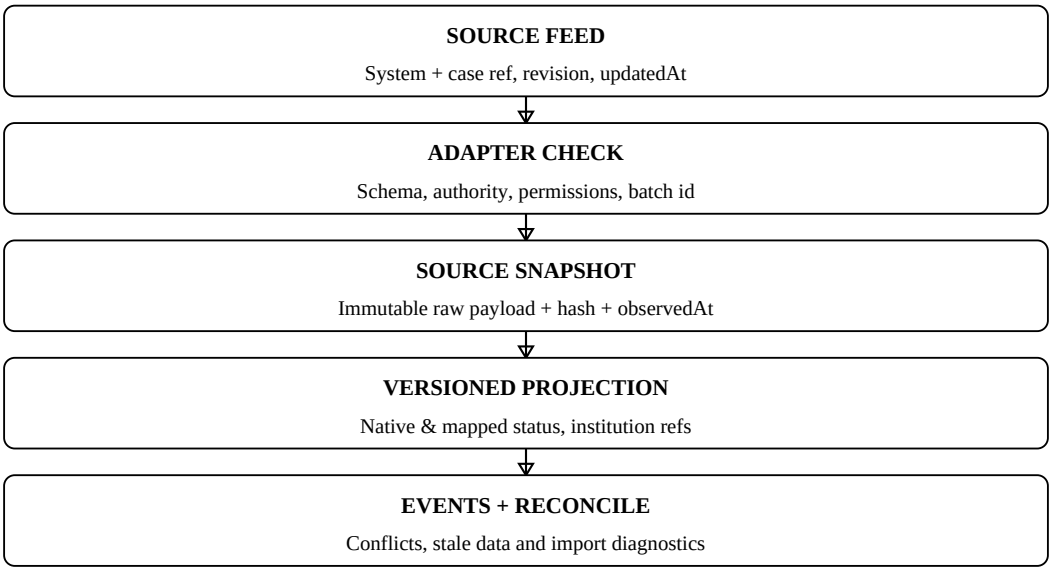


Figure: Every connector writes immutable source truth first; normalized state and workflow analytics are derived, with explicit conflicts and absent-feed conditions.

Adapter contract


```
{
  "sourceSystem":"RB-DEMO", "sourceCaseRef":"RB-007",
  "sourceRevision":"0007", "sourceUpdatedAt":"2026-10-21T09:20:00+05:30",
  "observedAt":"2026-10-21T09:30:01+05:30",
  "batchId":"SEED-20261023-1", "rawHash":"sha256:...",
  "nativeStatus":"Action Initiated", "fieldAuthority":{"status":"RB"},
  "classification":"RESTRICTED", "accessPolicy":"RB_POLICY_V1"
}
```


Integrity rules

• Current: fresh successful sync relative to approved freshness budget. Stale: watermark age exceeds threshold. Conflicted: authoritative sources disagree. Not connected: source absent - coverage unknown.

• When a late feed arrives, preserve official eventTime versus observedAt; UI displays both where necessary.

• Source field authority can differ: e.g. one bureau owns its native procedural status, not a global source overriding all other bureaus.

• Escalate mismatched Permanent IDs, invalid status mappings, missing source keys and duplicate revisions to data-health review.

• Never use data-ingest timing alone as evidence that an officer failed to act by deadline.

12. SLA, dashboard metrics and reports

Configured, not invented, SLA rules

Prescribed timelines depend on authoritative policy, grievance category, source and administrative procedure. DARPG CPGRAMS guidance is public context, not blanket proof of the governing timeline for each AICTE institution complaint. A policy admin must configure rules, start/pause/end events and holiday calendars with approval and effective dates.

Metric	Formula and disambiguation	Must display
Open/pending	Count cases mapped to PENDING or IN_PROCESS at selected as-of time	Case scope, source coverage, definition version

Metric	Formula and disambiguation	Must display
Transferred	Count source cases/events classified as transferred within window	Whether snapshot or event-count measure
Disposed	Count cases with authorized disposition event in selected window	Denominator and per-source status mapping
Disposal rate	Disposed cases / eligible cases in selected cohort/window	Cohort definition; do not mix stock and flow
Overdue	Open cases with active approved SLA rule and dueAt < reference clock	Timezone, pause, as-of date, unknown policy count
Pending age	Reference date minus receipt/effective event date per policy	Median, ageing buckets and censored incomplete cases
Overlap rate	Reviewer-confirmed related pairs / reviewed pairs	Not equivalent to misconduct or savings
Coordination success	Verified completed coordination records / accepted coordination records	Unfinished cases still counted separately

Report/filter catalogue

• Institution-wise history (all permitted cases and movement ledger); region-wise and bureau-wise workloads; status-wise totals.

• Date-range report by receipt date, action date or completion date, selected explicitly.

• Pending age buckets, near-due and breached policy clock, paused clock share, unresolved transfer acks.

• Monthly/quarterly disposal trend with source coverage and mapping version; overlap review and false-match counts.

• CSV/PDF print/role-filtered export; record file creation event, actor and applied filters.

Seeded SLA policies that actually trigger - SYNTHETIC DEMO ONLY

These values are fictional training defaults for local demo accounts. They are deliberately NOT described as AICTE rules. An official CPGRAMS 21-day guidance exists for CPGRAMS cases, but applicability to a specific AICTE Bureau/Cell case requires separate written mapping and approval. In production mode, demonstration policies cannot be selected: missing approved policy produces POLICY_NOT_CONFIGURED / UNKNOWN.

Policy ID	Start and target (Asia/Kolkata)	Near due / breach / escalation
DEMO-CASE-STANDARD-v1	receiptAt + 14 calendar days	near due <=72h; overdue when dueAt passed; escalate to Bureau Lead at +24h
DEMO-CASE-PRIORITY-v1	receiptAt + 3 calendar days	near due <=24h; overdue immediately; escalate at +12h
DEMO-TRANSFER-ACK-v1	transferRequestedAt + 2 business days	near due <=1 business day; bureau lead escalation +24h after due
DEMO-SYNC-ACK-v1	blindNoticeAt + 48 elapsed hours	near due <=12h; coordinator escalation +12h after due
DEMO-TASK-v1	task.assignedAt + 5 business days	near due <=1 business day; lead escalation +1 business day

Clock semantics and synthetic trigger examples

• Business calendar: Monday-Friday 09:00-17:00 Asia/Kolkata in the synthetic demo, with a versioned demonstration holiday list. Deadline addition for business-day rules moves to the same local clock after eligible days; test Friday-to-Tuesday transitions.

• For STANDARD, receipt 01 Oct 2026 10:00 IST yields due 15 Oct 2026 10:00 IST (14 calendar days). At 12 Oct 10:00 it becomes NEAR_DUE; at 15 Oct 10:00:01 it becomes OVERDUE; at 16 Oct 10:00:01 escalation is eligible. Test on a frozen reference clock.

• PAUSED applies only to an approved source/bureau-specific pause code (e.g., waiting for requested information) with audited pauseFrom and resumedAt. Add elapsed eligible pause duration to dueAt; no silent pauses or arbitrary officer edits.

• Case status DISPOSED pauses further open-case breach alerting but retains the historical breach record; transfer does not automatically restart the statutory clock.

• Scheduler persists deterministic alert key (policyId, caseId/taskId, state, thresholdAt, version). Reprocessing and time travel produce zero duplicate delivery intents.

External notifications: implemented contract, not a disabled afterthought

Channel	How it works	Safety/conditions
In-app (core)	MongoDB notification + websocket/poll; read/ack states persist	Full actionable info only within authorized login
Email (core integration)	MongoDB outbox worker to SMTP/provider (local test sink by default); templates + retries + bounce status	Send to approved officer work email only; subject: action needed, no case narrative/PII
SMS (deployment-gated)	Provider-neutral SMS transport and local simulator; actual gateway behind admin approval flag	Minimal opaque code + secure login link; explicit organization approval & sender requirements
Fallback / dead letter	Retry with backoff 1, 5, 30 min; cap attempts; dead-letter/admin repair	Never mark delivered until provider acknowledges; idempotent send key
Preferences / quiet hours	Urgency tier, escalation routing, officer channel preferences and timezone	Mandatory legal notifications require separately validated policy

Example payload - safe email/SMS notification

```
{ eventType: "OVERLAP_REVIEW_REQUIRED", notificationId: "NOT-0005",
  recipientOfficerId: "demo-rb-01", deliveryChannel: "EMAIL",
  subject: "NEXUS-IC: coordination review requested",
  body: "A restricted coordination item awaits your review. Sign in to NEXUS-IC.",
  safeLink: "/review/FND-004", idempotencyKey: "NOT-0005:EMAIL:v1",
  noCaseNarrative: true, sourcePolicy: "DEMO-ALERT-v1" }
```

Test with a local email sink; verify retry, duplicate suppression, bounce/dead-letter, recipient change and zero leaked complainant fields. Production SMS and email cannot be claimed live until transport and permissions are validated.

13. Privacy, security, reliability and accessibility

Area	Required implementation	Negative test / risk
Auth/session	HttpOnly Secure cookies or safely rotated sessions; server rechecks authorization	Token theft/localStorage exposure avoided
Cross-bureau privacy	Default-deny ABAC and opaque notices	Forbidden narrative absent from JSON/export/socket
Input and output	Schema validation, restricted attachments, content security protections	XSS/injection/untrusted file payload
Concurrency	Version lock and idempotency key for stateful operations	No duplicate task after network retry
Audit	Append-only events with actor, purpose, timestamp, before/after ref	Silent edits not permitted
At rest/in transit	Encrypted storage and HTTPS proposed for real deployment	No secret/data accidentally committed
Data minimization	Synthetic sample, scoped fields, purpose bound retention	No authentic complainant data in demo
Accessibility	Keyboard, skip link, field labels, WCAG 2.1 AA-informed review	200% zoom, focus order, screen-reader labels
Observability	Health probes, request IDs and non-sensitive structured logs	No complainant text in logs
Disaster readiness	Backup, restore, import replay, signed artifact digests	Recover from failed deploy/seed state

Government UX and information integrity

Design to GIGW 3.0 and WCAG 2.1 Level AA practices in the implementation, then independently test. “Accessible by design” is not a certification. Any production environment also needs AICTE approval for identity, security, retention, procurement and integration. [4]

14. Ten strategic improvements - complete technical specifications

These ten enhancements are ADDITIONS OR EXTENSIONS to baseline F01-F64, not replacements. Each entry cites the original features it extends; overlapping capability is intentionally reused rather than counted as a completely new system. “P0/P1/P2” denote planned implementation priority, not completion status.

ID	Strategic improvement	Mapped baseline feature IDs	Priority
E01	NEXUS PREVENT - Action Conflict Prevention	F17, F23, F24, F27, F48	P1 (core extension)
E02	NEXUS SYNC 2.0 - Accountable Coordination	F18-F22, F25-F27, F41, F42	P0 + P1
E03	Institution Evidence Knowledge Graph	F01-F06, F11, F14, F19	P1
E04	Explainable Multilingual Relationship Matching	F02, F03, F17-F19, F28, F62	P1 measured semantic ranking (rules fallback)
E05	Smart SLA and Bottleneck Intelligence	F10, F12, F22, F35, F39-F44	P0 policy + P1 analytics
E06	Privacy-Safe Cross-Bureau Collaboration	F20, F25, F45-F47, F54	P0 + P1
E07	Institution Attention Radar	F04, F30, F33, F35, F37, F39, F48	P1
E08	Data Trust, Field Authority and Reconciliation	F09, F15, F27, F48, F49, F53	P0 + P1
E09	NEXUS IMPACT - Measurable Coordination Value	F35, F36, F58, F59, F60	P1
E10	Interactive Case Command Centre	F04, F29-F34, F37, F50, F55, F56, F59	P0 + P1

E01 / NEXUS PREVENT - Action Conflict Prevention

Priority: P1 (core extension). Existing feature anchors: F17, F23, F24, F27, F48.

Problem addressed and value. Add an officer-facing pre-action workspace for document requests, inspections, notices and follow-ups. Compare the *proposed action* to active known actions, not just the complaint text. Avoid duplicate effort before it is created.

Implementation and data. Fields: verifiedInstitutionId, actionClass, purposeCode, documentClasses, periodStart/periodEnd, proceedingRef, ownerBureau, sourceCompleteness. Candidate eligibility requires verified identity plus compatible action purpose and overlapping temporal scope; contradictory proceeding evidence is visible.

API contract / technical deliverable. POST /api/v1/prevent/check -> decision CLEAR|REVIEW|UNKNOWN; reasons[], counterReasons[], candidateActionIds[], sourceCoverage, ruleVersion, evaluatedAt. Store immutable preflight_findings with unique idempotency key.

Live demonstration / acceptance. Judge drafts an RB document request already requested by PGRC; REVIEW appears. Changing the period or document class changes the output. Disconnecting PGRC returns UNKNOWN, not CLEAR.

Safety and scope control. No automatic statutory block. Reviewer may proceed with a reason or open NEXUS SYNC. A real gateway enforcement mode requires explicit AICTE legal authority, formal policy and independent testing.

E02 / NEXUS SYNC 2.0 - Accountable Coordination

Priority: P0 + P1. Existing feature anchors: F18-F22, F25-F27, F41, F42.

Problem addressed and value. Complete the six-step handshake with requests, acknowledgement, lead/support roles, dependencies, accept/decline, request-more-information, stalled handovers, disagreement escalation and verifiable resolution receipts.

Implementation and data. coordination state machine: PROPOSED -> NOTICE_SENT -> ACK_PENDING -> ACCEPTED/DECLINED/INFO_REQUESTED -> PLAN_ACTIVE -> EVIDENCE_REVIEW -> COMPLETED. A declined or expired request remains visible and auditable. Never transfer source-case ownership implicitly.

API contract / technical deliverable. POST /api/v1/coordination/:id/respond; POST /api/v1/coordination/:id/escalate; GET /api/v1/coordination/:id/receipt; all transitions require permitted actor, prior version, idempotency key and audit event.

Live demonstration / acceptance. RB sends privacy-safe coordination request, PGRC acknowledges with a separate login, lead assigns task, dependent task completes and a redacted receipt reconstructs the sequence.

Safety and scope control. Task completion is not case disposal; preserve separate RB and PGRC case statuses and legal proceedings.

E03 / Institution Evidence Knowledge Graph

Priority: P1. Existing feature anchors: F01-F06, F11, F14, F19.

Problem addressed and value. Show a browsable, institution-centered graph with institutions/campuses, source cases, bureaus, officers, actions, tasks, documents and typed *reviewed* relationships.

Implementation and data. Use MongoDB graph_nodes (projection) and case_relationships/graph_edges (typed source-backed assertions), indexed by institutionId/fromId/toId/type. Server constructs a bounded neighborhood, redacts hidden nodes and never infers identity transitively.

API contract / technical deliverable. GET /api/v1/graph/institutions/:id?depth=1&limit=75 -> nodes[], edges[], visibleScope, omittedCount, asOf. React Flow layout; edge click opens its evidence card and permitted source record.

Live demonstration / acceptance. Click institution -> RB case -> suspected matching PGRC case -> evidence card. An unrelated case is shown separate; a forbidden attachment or person node is omitted.

Safety and scope control. Graph edges are not proof of wrongdoing, and no link appears purely because two cases share a common neighbor.

E04 / Explainable Multilingual Relationship Matching

Priority: P1 measured semantic ranking (rules fallback). Existing feature anchors: F02, F03, F17-F19, F28, F62.

Problem addressed and value. Support differently phrased complaints in English, Hindi and other locally validated languages through verified-ID blocking, normalized keywords, date/entity signals and optionally a multilingual reranker.

Implementation and data. Persist original UTF-8 text, detected language, reviewer-correctable translation, input language and model/rule version. Ranking needs both positive clues and counter-signals such as distinct incidents, periods, campuses or proceedings.

API contract / technical deliverable. POST /api/v1/matching/candidates; POST /api/v1/matching/evaluate. Response includes candidate relation type, reasons with source pointers, missing data, evidence and modelVersion (nullable).

Live demonstration / acceptance. Two differently worded cross-language synthetic complaints rank as review candidates; same-PID unrelated cases remain low confidence or KEEP_SEPARATE.

Safety and scope control. Never present unmeasured language accuracy or auto-merge. Use locked institution-grouped holdout with precision/recall and language-stratified false positives.

E05 / Smart SLA and Bottleneck Intelligence

Priority: P0 policy + P1 analytics. Existing feature anchors: F10, F12, F22, F35, F39-F44.

Problem addressed and value. Provide due-soon/overdue calendars, handover delays, stalled coordination tasks, reasoned follow-up suggestions and bureau workload analysis from policy-approved clocks.

Implementation and data. Policy table defines applicable category, start event, target days, business calendar, pause/resume reasons, holidays, applicable bureau and effective version; keep source event time distinct from ingest observation time.

API contract / technical deliverable. GET /api/v1/sla/attention?scope=...; POST /api/v1/sla/policies/:id/versions. Suggested priority uses overdue days, unacknowledged transfer and verified missing information - not misconduct score.

Live demonstration / acceptance. Move an action date across a tested holiday/pause boundary; due date and reason trail recompute correctly without rewriting historical policy versions.

Safety and scope control. Predicted disposal dates only after valid local historical evaluation; no assumed AICTE-wide statutory SLA.

E06 / Privacy-Safe Cross-Bureau Collaboration

Priority: P0 + P1. Existing feature anchors: F20, F25, F45-F47, F54.

Problem addressed and value. A bureau initially sees a blind, opaque overlap notice. Restricted narratives, identifiers and evidence are disclosed only through purpose-bound reviewed grants.

Implementation and data. ABAC computes actor + bureau + case + category + purpose + field policy and approved grant with expiry. Apply to API, search, graph, Socket.IO, file downloads and exports. Log reveal/deny and policy change.

API contract / technical deliverable. POST /api/v1/access-requests; POST /api/v1/access-requests/:id/decision; GET /api/v1/disclosures/:id. Do not include hidden case titles or narrative in errors, count badges or notification payloads.

Live demonstration / acceptance. RB cannot read PGRC protected material before approval; reviewed grant permits only allowed fields and expires automatically. Export after expiry is denied.

Safety and scope control. No general admin bypass; approval policy and retention must be formally verified before a real pilot.

E07 / Institution Attention Radar

Priority: P1. Existing feature anchors: F04, F30, F33, F35, F37, F39, F48.

Problem addressed and value. Give managers an operational queue of institutions needing administrative follow-up because of verified overdue actions, stalled acknowledgements, unresolved source conflicts or repeated pending tasks.

Implementation and data. Compute transparent attention reasons (counts, due dates, conflict IDs, acknowledged backlog); link each flag to permissible evidence. No opaque institutional wrongdoing ranking, no automatic penalty or regulatory recommendation.

API contract / technical deliverable. GET /api/v1/attention/institutions?region=...; response reasons[], sourceCoverage, overdueTaskCount, conflictCount, taskOwner, dataAsOf.

Live demonstration / acceptance. Select a high-attention fictional institution, inspect linked overdue task and stale source. When feed disconnects, mark incomplete rather than ranking on fake zero.

Safety and scope control. Call it attention/operations prioritization; never institution guilt or compliance risk score.

E08 / Data Trust, Field Authority and Reconciliation

Priority: P0 + P1. Existing feature anchors: F09, F15, F27, F48, F49, F53.

Problem addressed and value. Expose data health and field-level disagreements, not only connector status. Quarantine malformed imports, support replay and retain every source-native status.

Implementation and data. Immutable source_snapshot carries system, version, watermark, source_updated_at, observed_at, payload hash and access label; source_claims carries field, assertedValue, issuing source and authority priority. Reconciler proposes attributed resolution.

API contract / technical deliverable. GET /api/v1/data-quality/claims; POST /api/v1/data-quality/conflicts/:id/resolve. Accepted correction creates new projection version and audit, never mutates raw snapshots.

Live demonstration / acceptance. Import two contradictory statuses, inspect both with timestamps, reconcile only an allowed field, rerun batch and show zero duplicated cases.

Safety and scope control. Any source not connected is UNKNOWN, not 0. A delayed feed does not prove an officer missed a deadline.

E09 / NEXUS IMPACT - Measurable Coordination Value

Priority: P1. Existing feature anchors: F35, F36, F58, F59, F60.

Problem addressed and value. Provide a reproducible comparison lab instead of claiming unmeasured savings: same seeded scenarios through conventional independent-bureau handling and through PREVENT + SYNC.

Implementation and data. Define operational metrics in advance: duplicate-request attempts identified, review precision, irrelevant alert share, acknowledgement latency, officer steps, unresolved handovers, case-age distribution. Label observed simulation vs measured pilot vs hypothesis.

API contract / technical deliverable. POST /api/v1/impact/scenarios/run (demo admin only); GET /api/v1/impact/trials/:id; GET /api/v1/impact/definitions. Store trialId, fixture hash, method, event log, start/end, outcome and comparison method.

Live demonstration / acceptance. Judge triggers identical seeded scenarios in both modes and sees an audit-backed comparison with raw event logs, denominators, caveats and reproducibility hash.

Safety and scope control. “Action flagged” does not equal “action avoided”; only trace-backed non-execution after documented coordination qualifies as observed avoided work in the prototype.

E10 / Interactive Case Command Centre

Priority: P0 + P1. Existing feature anchors: F04, F29-F34, F37, F50, F55, F56, F59.

Problem addressed and value. Build a role-personalized responsive interface: command KPIs, Institution 360, full case workspace, overlap review, PREVENT desk, SYNC board, graph, radar, source-trust view, Impact Lab and reports.

Implementation and data. React JSX components, React Router, TanStack Query, Tailwind, Recharts, React Flow. Every data panel has loading/empty/error/UNKNOWN states, scope/asOf metadata, keyboard focus and accessible controls.

API contract / technical deliverable. Routes /dashboard, /institutions/:id, /cases/:id, /review, /prevent, /sync, /graph/:id, /attention, /impact, /reports, /admin. Deep-links enforce auth and expose no hidden fields.

Live demonstration / acceptance. A reviewer demonstrates all major flows without dead-end buttons. At 200% zoom all critical actions remain reachable by keyboard, with no incorrect chart totals.

Safety and scope control. Graph and animations are optional presentation, never a substitute for reliable workflow logic or screen-reader information.

15. Unified architecture, new data models and execution contracts

The system remains a modular JavaScript monolith; feature-rich does not mean microservice-heavy. Deterministic rules run reliably without any generative AI service. An optional multilingual ranker is a bounded internal adapter, not an agent with administrative authority.

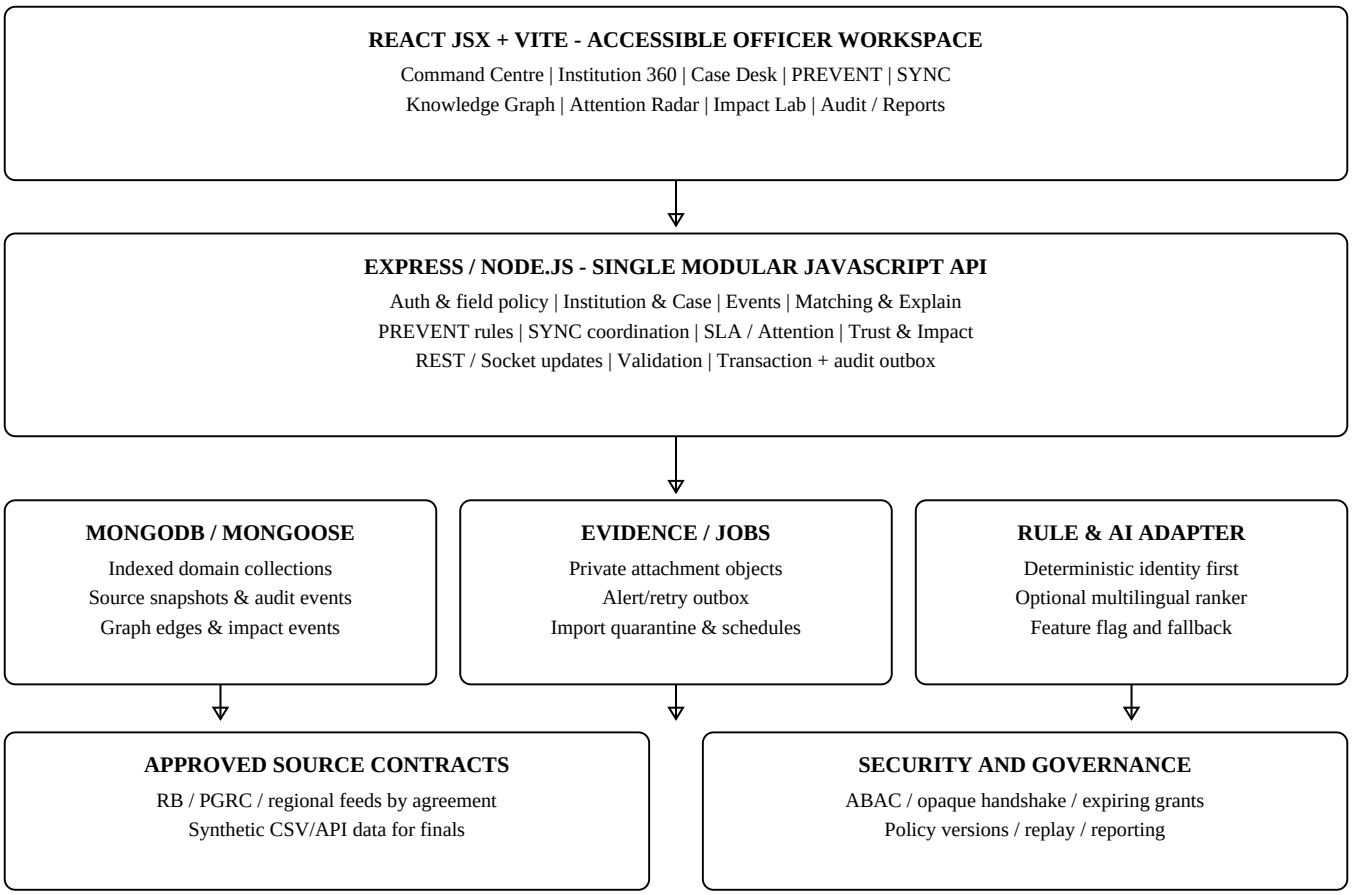
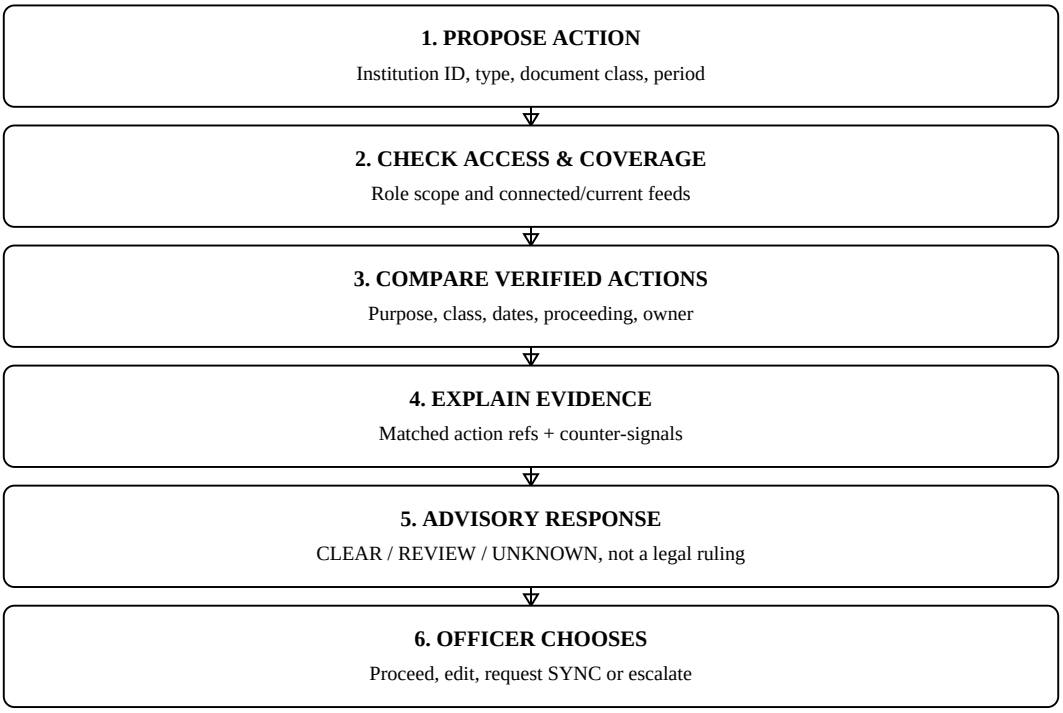


Figure: Updated deployment architecture integrating all 10 improvements with the original case platform, persistence, audit, source adapters and optional multilingual support.

NEXUS PREVENT - authoritative decision path



No AI-only hard block; only expressly approved gateway rules could prevent commits.

Figure: Proposed actions are compared before external administrative execution; advisory output does not block an official act.

Institution Evidence Knowledge Graph

Links are typed, reviewable, scoped and never transitively auto-merged.

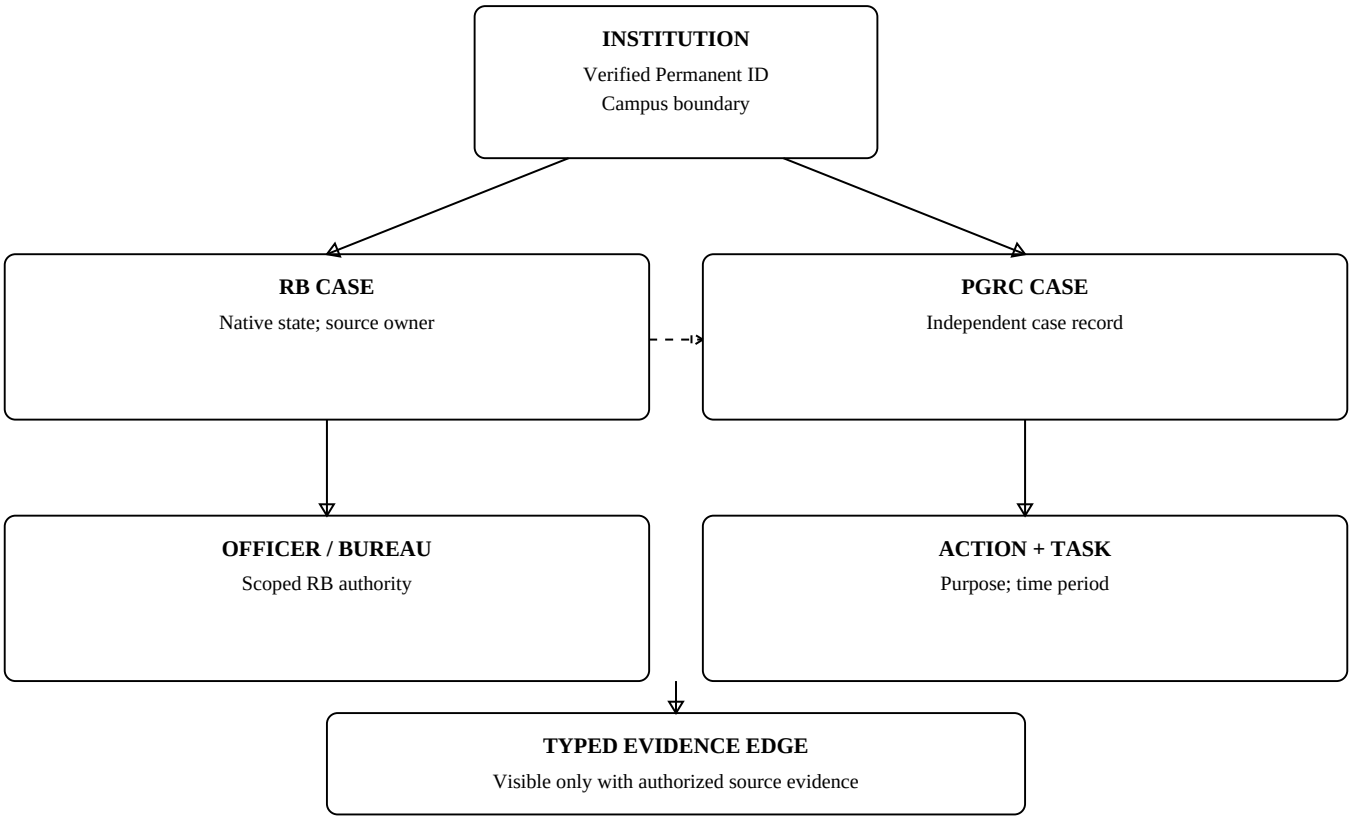


Figure: Typed relationship graph uses verified institution identity and permission-filtered nodes and edges.

Existing collections retained - new/extended collections

Collection / object	Essential stored fields	Invariant / index
action_intents	intentId, PID, actionClass, purposeCode, docClasses, period, actor, version	idempotencyKey unique; proposal not official action
preflight_findings	intentId, CLEAR/REVIEW/UNKNOWN, reasons, candidateIds, evaluatedAt, ruleVersion	immutable evaluation snapshot
graph_edges	fromId, toId, type, sourceRefs, visibility, decisionId, reviewer	typed edge; indexed by institutionId and source
attention_signals	institutionId, reasonCode, underlyingTaskIds, asOf, coverage, version	refreshable projection, no guilt score
impact_trials	fixtureHash, arm, runId, startedAt, scenarioVersion, environment	test-arm provenance, immutable raw traces
impact_events	trialId, actorRole, action, occurredAt, caseRef, taskRef, outcome	append-only, no sensitive narratives
field_claims	sourceSnapshotId, entityId, field, value, authority, observedAt	conflict preserved; reviewer resolves projection
access_requests	requester, case, fields, purpose, approver, grantedUntil	default deny, expired grant enforced
language_assertions	caseRef, detectedLanguage, originalHash, translationRef, reviewedBy	original never replaced
policy_versions	scope, calendar, startRule, stopRule, pauseRules, effectiveFrom	version immutable and traceable

Request and response design examples

```
POST /api/v1/prevent/check
{institutionId, actionClass, purposeCode, documentClasses, period, expectedVersion}
-> {decision:"REVIEW", ruleVersion, candidates:[opaqueId], reasons, counterReasons,
   dataCoverage:"CURRENT", evaluatedAt, preflightId}
```

```
GET /api/v1/graph/institutions/:id?depth=1&limit=75
-> {nodes:[{id,type,label,visibility}], edges:[{from,to,type,evidenceRef}],
   asOf, omittedCount, scope, sourceCoverage}
```

```
GET /api/v1/impact/trials/:trialId
-> {scenarioVersion, fixtureHash, baseline:{events,metrics}, nexus:{events,metrics},
   comparison:{definitions, denominators, limitations}, measurementType:"SYNTHETIC"}
```

Operational event order and consistency guarantees

1. Read user, bureau, purpose and field scope from verified backend session; never trust role in React local state.

2. Resolve exact verified Permanent ID, campus and source authority; if no trusted match, return identity-review/UNKNOWN.

3. For each proposed action, compare only role-permitted metadata and disclose the minimum reasons sufficient for action.

4. Commit coordinator decisions using expected-version + idempotency-key + case/coordination events in a Mongo transaction.

5. Write notification outbox within the transaction; deliver asynchronously; retry safely without duplicate notices.

6. Recalculate aggregate projections and attention reasons from authorized events; retain asOf and source coverage.

7. Record export and proof-trial provenance; action and metric results must be reproducible from persisted events.

Expanded frontend routes and components

Route	Primary components	Empty/denied/unknown behavior
/dashboard	KpiStrip, ScopedFilters, WorkloadCharts, FollowUpInbox	No source -> coverage banner, UNKNOWN count
/institutions/:id	IdentityHeader, CaseTimeline, SourceStatus, RelatedCases	Ambiguous campus -> review badge
/cases/:id	CaseHeader, NativeStatus, Events, MovementPanel, SecureEvidence	Unauthorized fields omitted by API
/review	CandidateEvidenceCard, CounterSignals, DispositionModal	Low-evidence candidate -> more information
/prevent	ActionIntentForm, PreflightDecision, KnownActions	Unconnected feed -> UNKNOWN, no green all-clear
/sync	HandshakeInbox, RoleAssignment, TaskDependencyBoard, Receipt	Declined/expired request is audited
/graph/:id	InstitutionGraph, EdgeDetail, EvidenceDrawer	Restricted node/edge never returned
/attention	FollowUpReasons, DueFilters, ConflictDrilldown	Missing scopes labelled incomplete
/impact	PairedScenarioRunner, MetricDefinitions, RawTraceViewer	No run -> no impact claim
/data-trust	SourceWatermarks, ConflictQueue, ImportErrors	No latest import -> explicit not connected
/reports	DateRange, MetricTable, ExportPanel, PrintableReport	CSV/API obey same ABAC filter

16. Technical UX, evaluation design and team skills

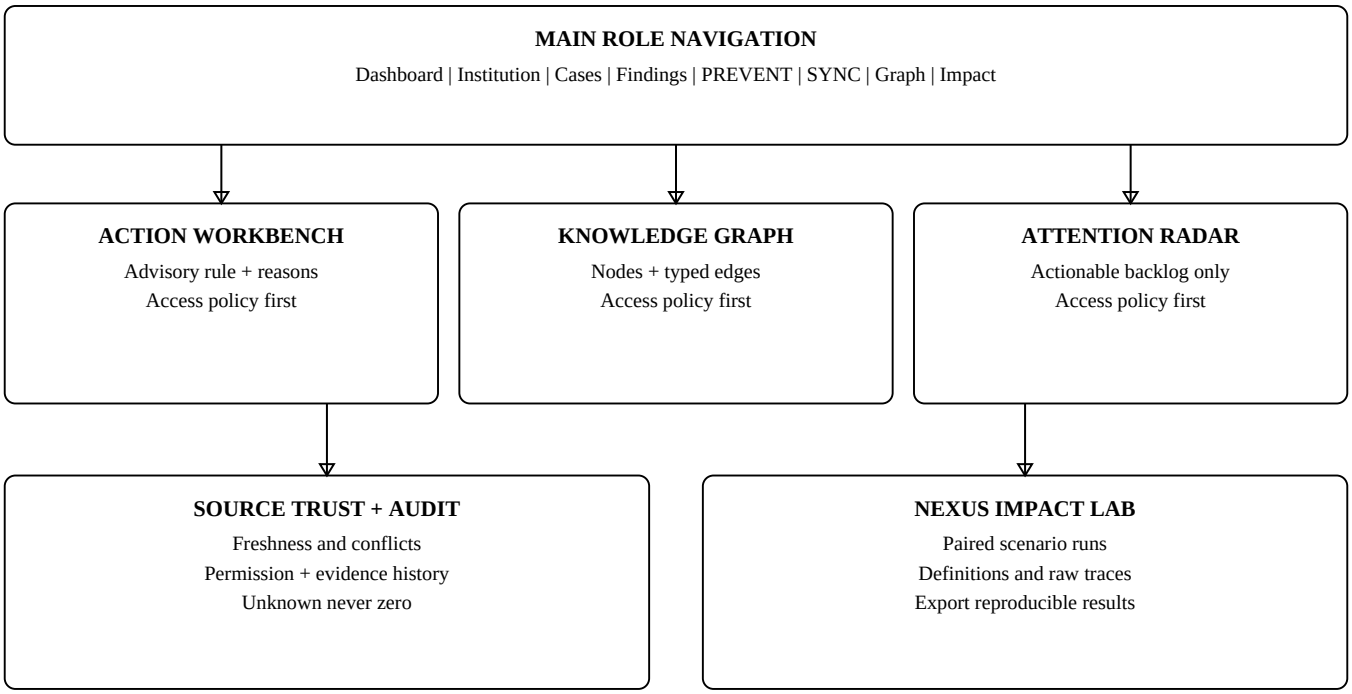
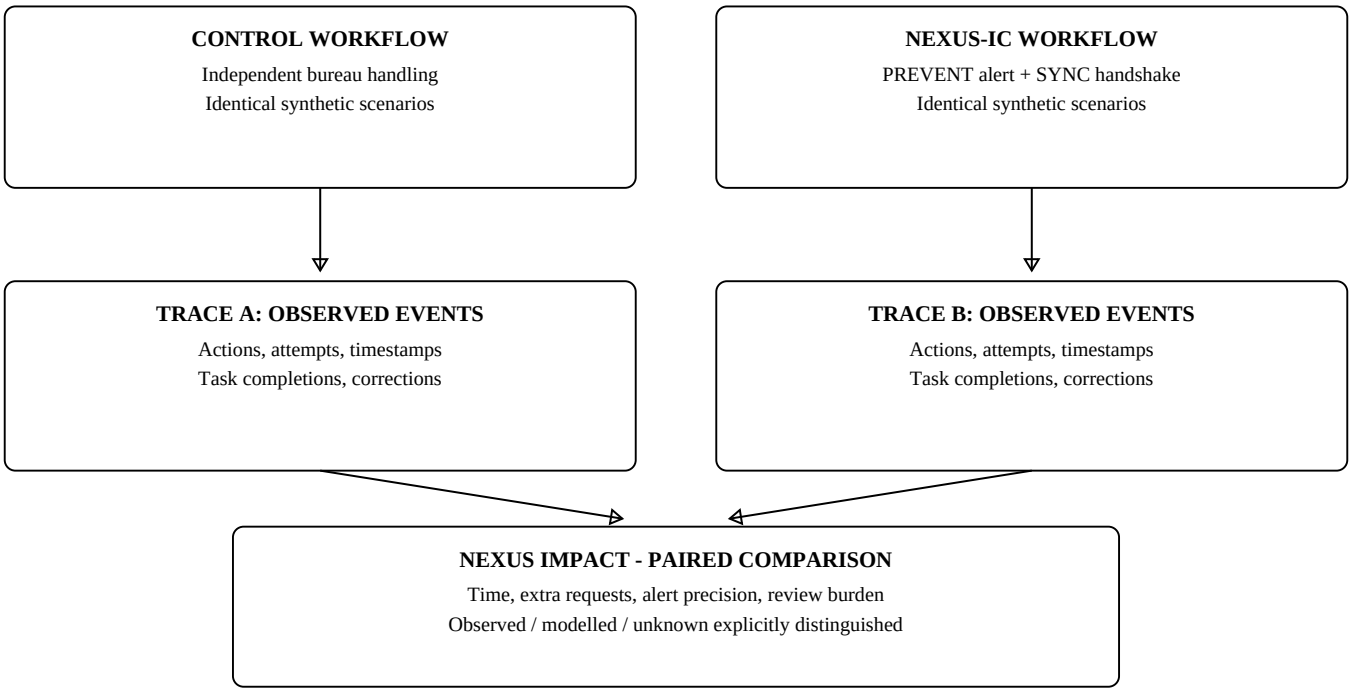


Figure: Unified navigation: the added PREVENT, Graph, Radar and Impact Lab screens integrate with original case and coordination journeys.

NEXUS IMPACT - defensible benchmark methodology



Simulated difference is not evidence of real AICTE savings.

Figure: Predefined, paired synthetic workflow runs; measured prototype effect is not equivalent to observed real-world AICTE efficiency gains.

Metric	Operational numerator and denominator	Failure to avoid
Alert precision	Confirmed relevant relation alerts / all alerts reviewed	Do not divide by all created cases
Unnecessary alert share	Reviewer-dismissed alerts / alerts reviewed	Report category and reviewer uncertainty
Duplicate action attempts	Action proposals matching an already active same-purpose+period action	An attempt detected is not saved work
Demonstrated avoided request	Confirmed coordinated request not repeated in completed scenario / eligible tasks	Must have event trace and exclusion rules
Acknowledgement latency	Accepted notice timestamp minus sent timestamp, by completed notice	Do not treat unacknowledged items as zero time
Review workload	Steps, time and correction count per scripted officer task	Compare identical scenario and role
Case ageing	Open cases by elapsed valid clock, scope and cohort	Do not infer disposal SLA without approved policy
Missing source fraction	Disconnected/stale required source streams / expected authorized streams	Not a complaint count or AICTE-wide denominator

Evaluation protocol - evidence standards

1. Freeze seeded case fixture set and scenario version before comparison; verify the same cases and scripted roles in each arm.

2. Define outcome labels independently; use a held-out institution-grouped set for matching and a separate hard-negative set.

3. Record raw event traces, effective date/time zone, tool versions, seed hash and machine environment.

4. Report per-language subgroup performance only if sample size is disclosed and adequate; do not claim generalization beyond synthetic labels.

5. Publish numerator, denominator and scope behind every displayed impact percentage; include confidence intervals only when statistical design justifies them.

6. Require privacy-negative and false-match suites to pass before enabling optional semantic suggestions.

7. Clearly separate measured synthetic-demo results from projected or hypothetical deployment benefits.

Detailed user journeys

Persona	Start -> decision -> outcome	Essential proof
RB officer	Login -> case search -> draft action -> PREVENT REVIEW -> request SYNC	Case does not silently transfer; result persisted
PGRC officer	Opaque notification -> ack -> request narrow field disclosure -> help coordinate	Unapproved narrative never exposed
Lead reviewer	Evidence card -> compare counter-signals -> record reason -> assign tasks	False positives can remain separate
Regional office	Filter institutions/region -> attention item -> pending handover	Only own delegated region visible
Senior management	View scoped workload and SLA trends -> inspect source health -> escalate	Every KPI reconciles with records
Independent auditor	Open coordination receipt -> replay relevant event trail -> verify export	Read-only scope, no silent edit

Development skills and ownership

Responsibility	Skills and concrete implementation ownership
UI / product	React JSX, Vite, forms, navigation, accessible components, Recharts, optional React Flow
Backend / workflow	Node.js, Express, JavaScript, request validation, REST, state machines, idempotency
Database / sources	MongoDB/Mongoose, compound indexes, transactions, projections, source import reconciliation
Matching / NLP	Identity blocking, rule features, text similarity, multilingual offline evaluation and explainability
Security / QA	RBAC/ABAC, threat modeling, Playwright, Supertest, negative-role tests, audit evidence
Design / jury	Problem-fit narrative, impact metric definitions, scenarios, repeatable demo, limitations

Role and scope in implementation

Source system owners must approve access, dataset classification, permanent-ID/campus semantics, field disclosure boundaries, applicable timelines and responsible escalation authority. A coding team cannot infer these from the challenge text alone. Government accessibility guidance (GIGW 3.0) and applicable Indian data-protection rules inform design; compliance requires a later institutional legal/security review. [4][5][6]

17. Test plan: 30 baseline acceptance checks

Test ID	Verifiable case	Category
T01	Exact verified same PID; correct bureau case linking	Positive
T02	Similar name but different verified PIDs	Hard negative
T03	Same institution, unrelated incident	Hard negative
T04	Related incident under legally separate proceedings	Hard negative
T05	Same document class but different periods	Hard negative
T06	Transitive A-B and B-C do not imply A-C	Hard negative
T07	Unknown source coverage is never displayed as zero	Safety
T08	Late-arriving source update preserves event vs observed time	Temporal
T09	Replaying same source batch creates zero duplicate cases	Import

Test ID	Verifiable case	Category
T10	Conflicting source statuses retained in disagreement queue	Integrity
T11	RB cannot export PGRC private narrative without grant	Security
T12	Opaque notification contains no private text	Security
T13	Expired disclosure grant stops reading docs	Security
T14	Role query/search/export/socket share same policy filter	Security
T15	Wrong actor cannot accept another bureau transfer	Authorization
T16	Review decision requires nonempty reason	Workflow
T17	Concurrent coordination updates reject stale version	Concurrency
T18	Action check returns UNKNOWN for missing feed	Safety
T19	Only configured rule marks SLA overdue	Correctness
T20	Policy version change does not rewrite historical ageing	Temporal
T21	Real-time count reconciles with filtered records	Data
T22	Bad CSV row quarantined; good rows imported	Import
T23	All essential flows keyboard navigable / 200% zoom	Accessibility
T24	Notification retry does not multiply alerts	Reliability
T25	Event correction does not erase original audit entry	Audit
T26	Synthetic end-to-end RB->PGRC->task->receipt persists	Judge demo
T27	Case relationship does not auto-close a source case	Safety
T28	Report export records who, filter, time and scope	Privacy
T29	Offline NLP service falls back to rule-only matching	Resilience
T30	Tampered audit hash is detected by verifier	Audit

Minimum evidence pack

1. Automated test logs with pass/fail, execution command, runtime and environment; never fabricate passing results.

2. Playwright screenshots/video of roles, two matching cases, a hard negative, decision, ack, task, audit and report.

3. Database fixtures and exact seed version hash; replay and reset scripts.

4. Measured latency/load results only when test actually run, with sample counts and machine specs.

5. Known limitations and unimplemented P1/P2 clearly flagged in readme and demo.

18. Risk register, open questions and definition of done

Risk / unknown	Mitigation or required owner decision
Official connector access unavailable	Synthetic adapters only; no live feed claims; API contracts require approval
Source ownership of status disputed	Keep native statuses; map analytically with field authority and review
One Permanent ID across campuses/years	Campus hierarchy and identity-review queue; test ambiguous cases
Cross-bureau privacy authorization	Field ABAC, blind notices, purpose-bound grants; obtain formal approval

Risk / unknown	Mitigation or required owner decision
Real SLA ambiguity	Configurable effective policies, never assert universal deadline
AI false positives	Hard-negative lock set, human dispositions and optional model release gate
Unapproved live email/SMS transport	SMTP email sink and SMS simulator available locally; live dispatch gated behind written approval
Poor internet / staging failure	Local Docker/Node demo and sealed seed/reset evidence
Large scope within one night	P0 path first; P1 only with end-to-end proof; P2 defer

Final acceptance gates

1. Every requirement R01-R55 is mapped and either verified working or explicitly listed as not yet implemented.

2. Every P0 feature F01-F64 tagged P0 passes its relevant UI and API tests; no role visibility leak.

3. Two-bureau judge scenario completes without manually editing the database.

4. Source history and actor provenance persist; missing feeds do not produce false zeros.

5. Real metrics come from actual local execution; no imaginary savings, accuracy, integration or deployment claims.

6. Readme contains environment setup, test commands, fixture reset, demo users, limitations, architecture and screenshots.

19. Additional enhancement verification: 22 release tests

These tests complement T01-T30. No feature is declared complete merely because it appears in this document. Every passed test must have evidence from an actual run.

Test	Verifiable behavior	Feature anchor
X01	PREVENT warns for verified same-PID, same-purpose, overlapping-period request	E01
X02	PREVENT clear for unrelated period or document class	E01
X03	Disconnected required source causes UNKNOWN not CLEAR	E01 E08
X04	PREVENT advice does not block legitimate official record updates	E01
X05	Declined and info-requested SYNC states preserve timeline and ownership	E02
X06	Two simultaneous ACK updates cannot overwrite one another	E02
X07	Coordination receipt reconstructs final tasks and countersigned notices	E02 E09
X08	Knowledge graph refuses inferred transitive A-C edge	E03
X09	Graph omits restricted nodes/edges for wrong role	E03 E06
X10	Multilingual matching shows references and counter-signals	E04
X11	Unrelated same institution remains a nonmatch across languages	E04
X12	SLA pause and holiday boundary produce correct versioned clock	E05
X13	Field grant expires and blocks files, search, graph and exports	E06
X14	Opaque alert response contains no protected names/narrative	E06
X15	Attention Radar flags only attributable work items, not guilt	E07
X16	Missing source degrades attention completeness visibly	E07 E08
X17	Source conflict remains visible after a reconciled projection change	E08

Test	Verifiable behavior	Feature anchor
X18	Impact run repeated with same fixture hash yields same deterministic events	E09
X19	Impact percent matches trace-based denominator; no action-found=avoided shortcut	E09
X20	Role-scoped command centre supports keyboard and 200% zoom	E10
X21	One browser journey spans PREVENT->SYNC->graph->impact without blank/error screens	E01 E02 E03 E09 E10
X22	Permission checks for API, realtime push and exports remain consistent	E06 E10

Proposed measurable gates (tune with mentors, not official SIH cutoffs)

Gate	Minimum completion proof	Blocking failure
Functionality	All R01-R55 mapped to working screen/API or clearly disclosed gap	Silent missing mandatory field / fake UI state
Privacy	No restricted narrative or node in negative-role API/download/export/socket	Any cross-bureau content leak
Identity	No false authoritative institution merge in locked negative tests	Same-name cross-PID merge
PREVENT	All known positive/hard-negative rules plus UNKNOWN pass on fixtures	Automatic arbitrary case/action block
SYNC	Ack/lead/task/receipt persist across role switch and reload	Unowned alert or silent reassignment
Data trust	Idempotent imports and source-conflict preservation work	Missing feed displayed as zero
Graph	Visible evidence ref per relationship; nontransitive matching	Unsupported or forbidden edge
Impact	Seed, definitions, event trace and denominators shipped together	Fabricated savings or misleading percentages
Accessibility	Keyboard, focus, labels and 200% zoom on core and added journeys	Critical task unavailable without mouse

20. Unified delivery roadmap and project definition of done

The ten improvements enrich the complete target system, but should be delivered behind functional gates. This is the only canonical consolidated roadmap and takes precedence over any older blueprint. All milestones and demo tasks are owned here; there is no competing schedule.

Release stage	Integrated feature coverage	Strict completion evidence
S0 - foundation	Repo and seed, React JSX, Express, MongoDB, authentication, basic UI state	Reproducible install and reset; both demo roles
S1 - mandatory PS	F01-F13, F29-F35, F39-F42, F45-F50, F55-F58	All eight PS groups demonstrable; all field/filter/status requirements
S2 - trust and relationships	F14-F19, F48-F49, E04 baseline, E08	Hard negatives, preserved snapshots, idempotent replay, case evidence
S3 - NEXUS SYNC 2.0	F20-F22, F25-F27, E02 and E06	Full cross-bureau handshake, field denial, owner task, receipt
S4 - NEXUS PREVENT	F23-F24, E01 + versioned action rules	REVIEW/CLEAR/UNKNOWN with persisted reasons and no AI-only blocks
S5 - finalist differentiation	F05-F06, F28, F36-F38, F43-F44, E03/E04/E05/E07/E09/E10	Permission-filtered graph, radar, Impact Lab, accessible experience

Release stage	Integrated feature coverage	Strict completion evidence
S6 - advanced optional	F51-F54, F59-F64, pilot adapters; F28 moved to measured P1 target	Feature flags; language tests; institutional approvals
S7 - release evidence	T01-T30 + X01-X22 + D01-D26, E2E demo, diagrams, screenshots, known limitations	No release with critical privacy, wrong identity or false result

Six-minute finals demonstration - improved integrated version

Clock	Live system action	Judge takeaway
00:00-00:35	RB role opens scoped dashboard and shows missing-source indicator	Not merely a generic public grievance form
00:35-01:10	Institution 360 + graph show distinct RB and PGRC cases	One verified institution; source cases remain separate
01:10-01:50	Explain differently phrased cross-bureau candidate and one rejected false positive	Evidence + contradictions, not opaque similarity
01:50-02:40	Draft document request; PREVENT returns REVIEW with underlying active action	Proactive prevention before duplicate execution
02:40-03:45	Request SYNC; switch PGRC role, accept and name lead with assigned task	Actionable, privacy-safe, persistent coordination
03:45-04:30	Complete task, show receipt, negative-role evidence denial	Real accountability and least-privilege enforcement
04:30-05:20	Open attention queue, SLA reason and reconciled data source view	Management insight with trusted numbers
05:20-06:00	Run Impact Lab paired scenario, show raw counts and repeatable reset	Proof-oriented value, no invented savings

Final explicit scope and exclusions

• All 64 baseline feature IDs F01-F64 remain specified; every strategic improvement E01-E10 is integrated and linked to its baseline dependencies.

• No Digital Twin is proposed or represented as built; NEXUS PREVENT is an explainable pre-action rules engine.

• No automatic case closure, guilt determination, unauthorized complainant-data sharing or statutory order generation.

• No claim of live AICTE databases, CPGRAMS write integration, government approval or demonstrated real-world savings.

• Optional multilingual ML and external connectors must remain correctly labelled until tested and authorized.

• Implementation is complete only when a test report references the exact code revision, environment, seed hash and observed pass/fail for each promised feature.

Source and claim policy

The problem-statement requirements and initial solution lineage derive from the user-supplied PS 26264 and NEXUS-IC 8.0. Official public context [4]-[6] informs government-friendly accessibility, grievance integration practices and privacy planning. Features E01-E10 are original design proposals; none is asserted as current AICTE operation or a measured improvement.

21. Nine-gap remediation checklist and final release decision

All nine critique points are corrected at specification level in this edition. The following D-series tests are executable acceptance requirements; none should be reported as PASS without actual code execution and archived evidence.

Test ID	Required acceptance evidence
D01	Seed DEMO-CASE-STANDARD-v1 from clean reset; scheduler computes dueAt receipt +14 calendar days
D02	Near-due alert appears within 72 hours and not before; clock frozen in test

Test ID	Required acceptance evidence
D03	Overdue and escalation fire after respective thresholds; no duplicate notifications on retry
D04	Business-day calendar and approved pause add time; source event vs observed time retained
D05	Production source without approved SLA shows POLICY_NOT_CONFIGURED, never demo defaults
D06	Verified distinct Permanent IDs block authoritative identity merge regardless of description
D07	RB and PGRC differently phrased positive fixture produces transparent computed 93.75 score
D08	Same-ID unrelated fire-door and certificate cases stay below alert threshold
D09	Threshold 60-79 shows exact Potential Overlapping Cases – Coordination Required text
D10	NLP outage displays RULE-ONLY and documented fallback, never invented semantic precision
D11	Evidence cards expose 7 contributions, missing fields, counter-signals and version
D12	Reviewer cannot coordinate without reason; saved KEEP_SEPARATE suppresses repeat alert
D13	Exact banner is accessible on Dashboard/Institution/Review, role-sensitive and timestamped
D14	Competent Authority has dedicated /authority route and delegated approval inbox
D15	Authority approval enforces requester != approver; audit attributes decision
D16	Authority cannot bypass field-ABAC to read private narratives or attachments
D17	External email delivered to sink via outbox; retry and idempotency trace present
D18	Email and SMS templates contain no complaint narrative or complainant PII
D19	SMS provider flagged off in sandbox; simulator receives the same minimal message
D20	RB authorized officer can discover PGRC case existence from unified case catalog
D21	RB without grant cannot download PGRC private files or case narrative
D22	Authorized temporary grant enables only selected fields until expiry; cross-role counts do not leak
D23	Search / graph / exports apply identical scopes and UNKNOWN source state
D24	Six wireframe-specified screens navigable and accessible at 200% zoom
D25	Before/after scenarios reproduce with same seed and report only trace-proven avoided actions
D26	PDF has unique sequential numbered main headings and one canonical roadmap/test/demo

Implementation-ready file and folder outline (all JavaScript / JSX)

```
nexus-ic/
  client/src/pages/{Dashboard,Institution,Case,Review,Prevent,Sync,Authority}.jsx
  client/src/pages/{KnowledgeGraph,Impact,Reports,DataHealth}.jsx
  client/src/components/{OverlapBanner,SourceCoverage,CaseTimeline}.jsx
  client/src/lib/api.js
  server/src/modules/{institutions,cases,matching,prevent,sync}/
  server/src/modules/{sla,notifications,authority,trust,impact}/
  server/src/policies/{catalogVisibility,fieldDisclosure}.js
  server/src/jobs/{slaAlerts,notificationOutbox,sourceSync}.js
  server/src/db/models/*.js
  server/src/seeds/{institutions,cases,roles,policies}.js
  tests/{api,matching,e2e,security,fixtures}/
  docs/{architecture,policy,modelCard,benchmark,knownLimitations}/
```

Reference implementation logic - JavaScript (illustrative, requires integration tests)

This is a precise JavaScript implementation sketch for the key relationship gate and scorer. In a real application the seven signal values are produced by versioned extractors with source references and negative tests; this code is not a substitute for a running server.

```
// server/src/modules/matching/scorePair.js
const WEIGHTS = Object.freeze({ issue:25, period:15, category:10,
  subject:15, remedy:10, entities:10, semantic:15 });
function scorePair(a, b, signal) {
  if (a.verifiedPid && b.verifiedPid && a.verifiedPid !== b.verifiedPid)
    return { state:"IDENTITY_BLOCK", score:null, reasons:["DIFFERENT_PID"] };
  if (!a.verifiedPid || !b.verifiedPid || !a.campusResolved || !b.campusResolved)
    return { state:"IDENTITY_REVIEW", score:null, reasons:["UNVERIFIED_IDENTITY"] };
  const contributions = Object.fromEntries(Object.entries(WEIGHTS).map(([key, w]) => {
    const n = Number.isFinite(signal[key]) ? Math.max(0, Math.min(1, signal[key])) : 0;
    return [key, Math.round(w*n*100)/100];
  }));
  const score = Object.values(contributions).reduce((s,n)=>s+n,0);
  const independent = ["issue","period","category","subject","remedy","entities"]
    .filter(key => (signal[key] ?? 0) >= 0.5).length;
  const bucket = score >= 80 && independent >= 3 ? "HIGH_REVIEW"
    : score >= 60 ? "REVIEW" : score >= 40 ? "LOW" : "NO_ALERT";
  return { state:bucket, score, contributions,
    missing:Object.keys(WEIGHTS).filter(k => !Number.isFinite(signal[k])),
    ruleVersion:"DEMO-MATCH-v1", requiresHumanDecision:true };
}
```

Reference demo SLA event logic

```
// server/src/modules/sla/classify.js; calendar-day policy only
function classifyDemoCase({receivedAt, now, status, policy}) {
  if (!policy || policy.approvedFor !== "SYNTHETIC_DEMO")
    return { state:"POLICY_NOT_CONFIGURED" };
  if (!receivedAt) return { state:"UNKNOWN_EVENT_TIME" };
  const dueAt = new Date(new Date(receivedAt).getTime()
    + policy.calendarDays * 86_400_000);
  const remainingMs = dueAt.getTime() - new Date(now).getTime();
  if (status === "DISPOSED") return { state:"CLOSED", dueAt };
  if (remainingMs < 0) return { state:"OVERDUE", dueAt };
  if (remainingMs <= policy.nearDueHours*3_600_000)
    return { state:"NEAR_DUE", dueAt };
  return { state:"ON_TRACK", dueAt };
}
const demoStandard = { id:"DEMO-CASE-STANDARD-v1",
  approvedFor:"SYNTHETIC_DEMO", calendarDays:14, nearDueHours:72 };
// Business-day or paused clocks use the separately specified calendar engine.
```

These snippets are implementation aids only; they do not cover approved pauses, holidays, PII-safe feature extraction or real government policies. Test the actual API, database events, outbound queue, calendar engine and role restrictions before release.

Final source-control and delivery gates

• Implement P0 items first; no P1/P2 feature is described as live until routed, persisted and tested. All 64 baseline IDs and 10 enhancements remain accounted for.

• The definitive acceptance suite is T01-T30, X01-X22 and D01-D26; run results, known failures, environments, seed hash and code revision must accompany an actual delivery.

• Before a government pilot: validate official bureau access policies, legal basis and processing agreements, statutory SLA definitions, official login integration, source API contracts, retention and transport permissions.

• No Digital Twin. No TypeScript. No automatic case merges or source closures. No claims of real AICTE access or quantified savings from fictional fixtures.

REFERENCES. Sources, assumptions and evidence boundary

[P] User-supplied PS 26264 text in this conversation - Primary requirements; not an externally validated legal interpretation.

[B] User-uploaded NEXUS-IC 8.0 Complete MERN Solution Document (22 October 2026) - Structural template and initial domain design; this edition changes TypeScript to React JSX and removes Digital Twin.

[1] AICTE Public Grievance Redressal Cell - https://www.aicte-india.org/bureaus/grievance-redressal

[2] AICTE Centralized Support System - https://www.aicte.gov.in/centralized-support-system

[3] CPGRAMS public portal, DARPG - https://pgportal.gov.in/

[4] Guidelines for Indian Government Websites and Apps, GIGW 3.0 - https://guidelines.india.gov.in/guidelines/

[5] DARPG comprehensive public grievance guidelines, 23 August 2024 - https://www.darpg.gov.in/sites/default/files/Comprehensive_guidelines_for_handling_the_Public_Grievances.pdf

[6] MeitY DPDP Rules, 2025 (check applicable enforcement stage and counsel) - https://www.meity.gov.in/documents/act-and-policies/digital-personal-data-protection-rules-2025-gDOxUjMtQWa